Assistant Professor
Department of Applied Mathematics
University of Dhaka
Dhaka-1000, Bangladesh

(2) Direction cosines of a straight line:

If α, β, γ are the angles that a given directed line make with the positive directions $X'OX$, $Y'OY$, $Z'OZ$ of the coordinate axes, then $\cos\alpha$, $\cos\beta$, $\cos\gamma$ are defined as the direction cosines of the line.

The direction cosines are usually denoted by l, m, n.

Let $A'OA$ be a line through the origin O and let it makes angles α, β, γ with the positive directions of the axes of coordinates. Then its direction cosines l, m, n are given by

$$l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma \qquad \text{——— (1)}$$

Let $P(x, y, z)$ be any point on this line and let OP has measured by r.

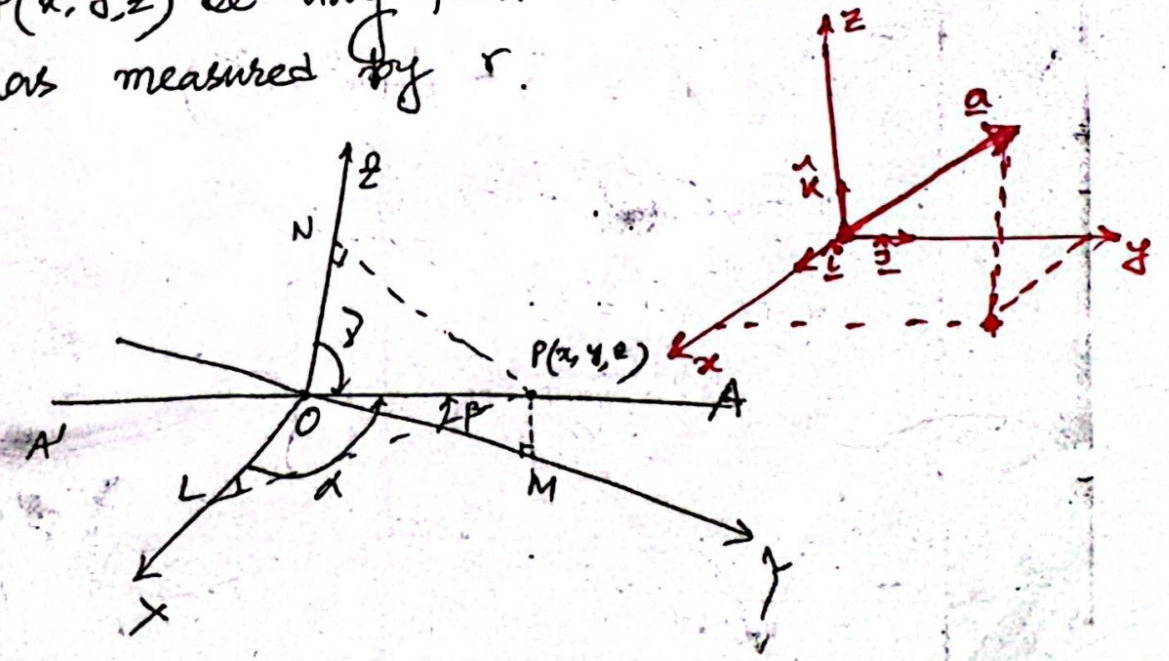
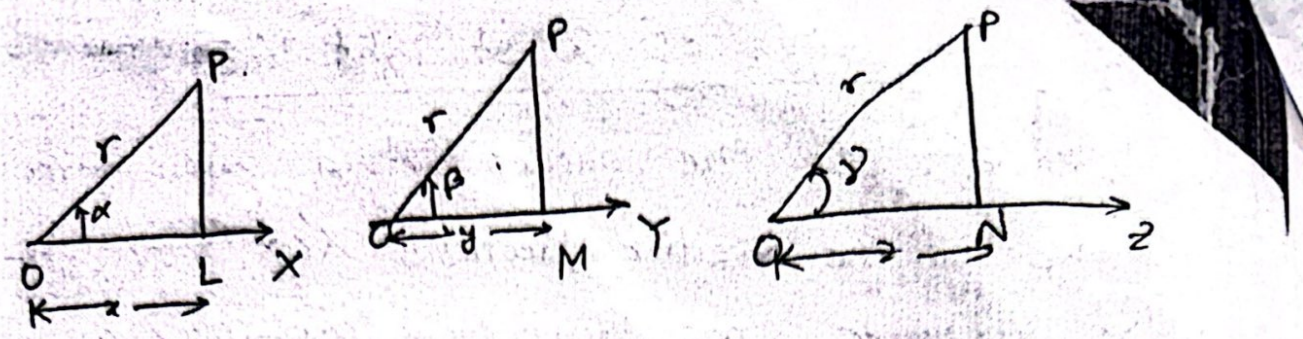


Fig - 1

CamScanner
Scanned with CamScanner



Draw PL, PM, PN perpendicular on OX, OY, and OZ, respectively.

Then $OL = x$, $OM = y$ & $ON = z$

Therefore, $OL = OP\cos\alpha \Rightarrow x = r\cos\alpha$

$OM = OP\cos\beta \Rightarrow y = r\cos\beta$

$ON = OP\cos\gamma \Rightarrow z = r\cos\gamma$

or, $x = lr$, $y = mr$, $z = nr$ ——— (2)

$\therefore x^2 + y^2 + z^2 = r^2(l^2 + m^2 + n^2)$

$\Rightarrow r^2 = r^2(l^2 + m^2 + n^2)$

$\Rightarrow l^2 + m^2 + n^2 = 1$

$\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ ——— (3)

Again, $\frac{x}{\cos\alpha} = \frac{y}{\cos\beta} = \frac{z}{\cos\gamma} = r$

$\Rightarrow \frac{x}{l} = \frac{y}{m} = \frac{z}{n} = r$ ——— (4)

CamScanner

Scanned with CamScanner

Remember relation (2) or (4) very carefully. These give the coordinates x, y, z of a point P in terms of its distance $OP = r$ from the origin & the direction cosines l, m, n of the line OP.

Note:

(i) The direction cosines of the x-axis are $(1, 0, 0)$

(ii) The direction cosines of the y-axis are $(0, 1, 0)$

(iii) The direction cosines of the z-axis are $(0, 0, 1)$.

Let a, b, c are given proportionals to the direction cosines of a line. Then

$$\frac{\cos\alpha}{a} = \frac{\cos\beta}{b} = \frac{\cos\gamma}{c} = \frac{\sqrt{\cos^2\alpha + \cos^2\beta + \cos^2\gamma}}{\sqrt{a^2+b^2+c^2}}$$

$$\Rightarrow \frac{\cos\alpha}{a} = \frac{\cos\beta}{b} = \frac{\cos\gamma}{c} = \pm \frac{1}{\sqrt{a^2+b^2+c^2}}$$

$$\therefore \cos\alpha = \pm \frac{a}{\sqrt{a^2+b^2+c^2}}$$

$$\cos\beta = \pm \frac{b}{\sqrt{a^2+b^2+c^2}}$$

$$\cos\gamma = \pm \frac{c}{\sqrt{a^2+b^2+c^2}} \quad \text{— (5)}$$

CamScanner

Scanned with CamScanner

The (+ve) or the (-ve) sign being taken according as the line is in the positive direction or in the negative direction.

Here, the numbers a, b, c proportional to the direction cosines are usually termed as direction-ratios.

(1) Direction cosines & direction ratios (Def^n):

The cosines of angles, which any given line makes with the positive directions of coordinates axes, are termed as the direction cosines of the straight line and the angles are called the direction angles.

That means, If α, β, γ are the angles that the straight line makes with the positive directions of the axes X, Y, and Z respectively, then $\cos\alpha, \cos\beta, \cos\gamma$ are called direction cosines of the line. DCs are generally denoted by l, m, n.

If three numbers are proportional to the direction cosines of any given straight line then the

CamScanner

Scanned with CamScanner

three numbers are called direction ratios of the straight line. It is generally denoted by a, b, c. If l, m, n be the dcs and a, b, c are the drs of a given straight line, then we have

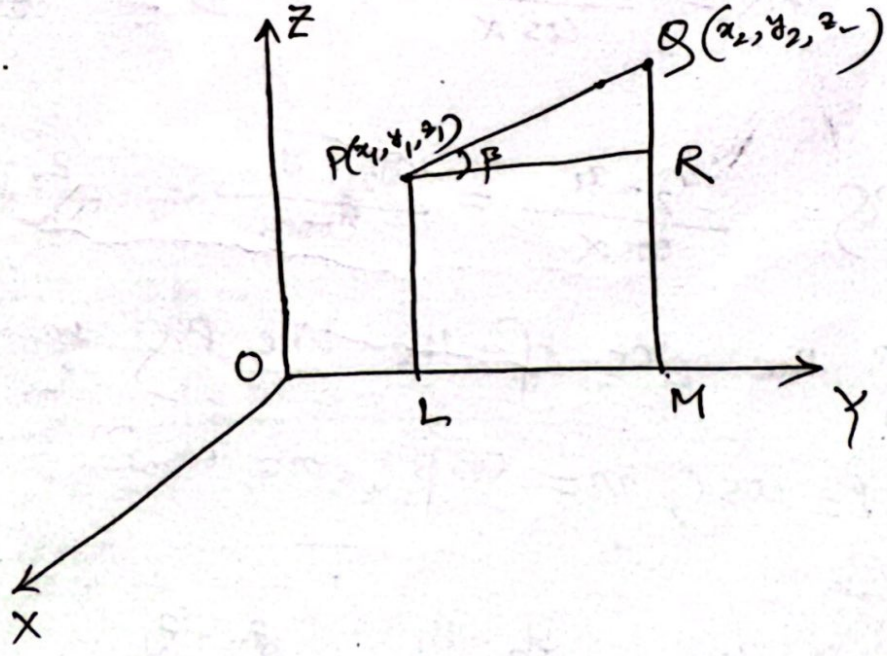
$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

Th^m:

If the coordinates of two points P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2) then show that the direction ratios of line PQ are

$$x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1 \quad \text{or} \quad x_1 - x_2,\ y_1 - y_2,\ z_1 - z_2$$

Proof:



Draw PL and QM perpendicular from P & Q point on Y-axis. Also draw PR ⊥ QM

From the figure,

$$LM = PR = y_2 - y_1$$

Let the line PQ makes an angle α, β, γ with X, Y and Z axis's, respectively. Then $\angle QPR = \beta$

From ΔPQR, $\cos\beta = \frac{PR}{PQ}$

$$\Rightarrow \cos\beta = \frac{y_2 - y_1}{PQ}$$

$$\Rightarrow PQ = \frac{y_2 - y_1}{\cos\beta}$$

Similarly, drawing perpendiculars on X & Z axes we can show

$$PQ = \frac{x_2 - x_1}{\cos\alpha} \quad \& \quad PQ = \frac{z_2 - z_1}{\cos\gamma}$$

Thus,

$$PQ = \frac{x_2 - x_1}{\cos\alpha} = \frac{y_2 - y_1}{\cos\beta} = \frac{z_2 - z_1}{\cos\gamma} \quad \text{---} (1)$$

Also, if the dcs of the line PQ are l, m, n then

$$l = \cos\alpha, \; m = \cos\beta, \; n = \cos\gamma$$

$(1) \Rightarrow$

$$\frac{x_2 - x_1}{l} = \frac{y_2 - y_1}{m} = \frac{z_2 - z_1}{n}$$

$$\Rightarrow \frac{l}{x_2 - x_1} = \frac{m}{y_2 - y_1} = \frac{n}{z_2 - z_1}$$

Scanned with CamScanner

$$\text{or, } \frac{l}{x_1-x_2} = \frac{m}{y_1-y_2} = \frac{n}{z_1-z_2}$$

We know that if a, b, c are the drs of the given line PQ, then

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

Thus, $a = x_1-x_2$ or x_2-x_1

$b = y_1-y_2$ or y_2-y_1

$c = z_1-z_2$ or z_2-z_1

Proved.

Corollary:

We know that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\Rightarrow 1-\sin^2\alpha + 1-\sin^2\beta + 1-\sin^2\gamma = 1$$

$$\Rightarrow \boxed{\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2}$$

Scanned with CamScanner

(A) **Problem:**

Find the dc's of a line that makes equal angles with the axes.

Sol^n:

Let l, m, n be the required dc's. Since the line makes equal angles with the axes, so we have

$$\alpha = \beta = \gamma$$

$$\Rightarrow \cos\alpha = \cos\beta = \cos\gamma$$

$$\therefore l = m = n \quad \text{——————} (1)$$

Also, we know that

$$l^2 + m^2 + n^2 = 1$$

$$\Rightarrow l^2 + l^2 + l^2 = 1 \quad [\text{using } (1)]$$

$$\Rightarrow 3l^2 = 1$$

$$\Rightarrow l^2 = 1/3 \Rightarrow l = \pm \frac{1}{\sqrt{3}}$$

Similarly,

$$m = \pm \frac{1}{\sqrt{3}}$$

& $$n = \pm \frac{1}{\sqrt{3}}$$

The required dc's are

$$\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$$

———o———

Scanned with CamScanner

Note: Here four such lines are possible, and their direction cosines are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$; $-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$; $\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ and $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$ as l, m, n and $-l, -m, -$ are the direction cosines of the same line.

~~Problem:~~

(Consider the fig-1 in 1st page)

Corollary:

If (x_1, y_1, z_1) be a point on the line $A'OA$, then from (2)

$$\frac{x_1}{\cos\alpha} = \frac{y_1}{\cos\beta} = \frac{z_1}{\cos\gamma} \quad \text{——— (*)}$$

$$\therefore \cos\alpha : \cos\beta : \cos\gamma = x_1 : y_1 : z_1$$

∴ If a line passes through the origin and the given point (x_1, y_1, z_1), its direction cosines are proportional to x_1, y_1, z_1.

Corollary:

Let a line whose d.c's are $\cos\alpha, \cos\beta, \cos\gamma$ passes through two given points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$.

Scanned with CamScanner

Transfer the origin to $P(x_1, y_1, z_1)$ and suppose that the new coordinates of Q are (x', y', z').

Then $x' = x_2 - x_1$, $y' = y_2 - y_1$, $z' = z_2 - z_1$.

Since the line passes through the point (x', y', z') and the new origin, we have

$$\frac{x'}{\cos\alpha} = \frac{y'}{\cos\beta} = \frac{z'}{\cos\gamma}$$

$$\Rightarrow \frac{x_2 - x_1}{\cos\alpha} = \frac{y_2 - y_1}{\cos\beta} = \frac{z_2 - z_1}{\cos\gamma} \quad \text{---------- *}$$

or, conversely, the d.c.s of the straight line passing through (x_1, y_1, z_1), (x_2, y_2, z_2) are proportional to $(x_2 - x_1) : (y_2 - y_1) : (z_2 - z_1)$

(Already shown in previous problem)

Assistant Prof.
Department of Applied Mathematics
University of Dhaka
Dhaka-1000, Bangladesh

Problem:

Given, $P(1,-5,7)$ and $Q(-3,6,-2)$ are two point
Find the direction cosines of OP, OQ and PQ.

Sol^n:

Let l_1, m_1, n_1 be the dc^s of OP

Then, $l_1 : m_1 : n_1 = 1 : -5 : 7$ $(\equiv a_1 : b_1 : c_1)$

Now, $a_1^2 + b_1^2 + c_1^2 = (1)^2 + (-5)^2 + (7)^2 = 75$

$$\therefore l_1 = \frac{1}{\sqrt{75}}, \quad m_1 = \frac{-5}{\sqrt{75}}, \quad n_1 = \frac{7}{\sqrt{75}}$$

In the same Day, the dc^s of OQ are found to be

$$l_2 = \frac{-3}{7}, \quad m_2 = \frac{6}{7}, \quad n_2 = \frac{-2}{7}$$

Let l, m, n be the dc^s of PQ

$\Rightarrow l : m : n = a : b : c = (x_2 - x_1) : (y_2 - y_1) : (z_2 - z_1)$

$\therefore l : m : n = (-3-1) : (6+5) : (-2-7)$

$= -4 : 11 : -9$

Since, $a^2 + b^2 + c^2 = (-4)^2 + (11)^2 + (-9)^2 = 218$, we have

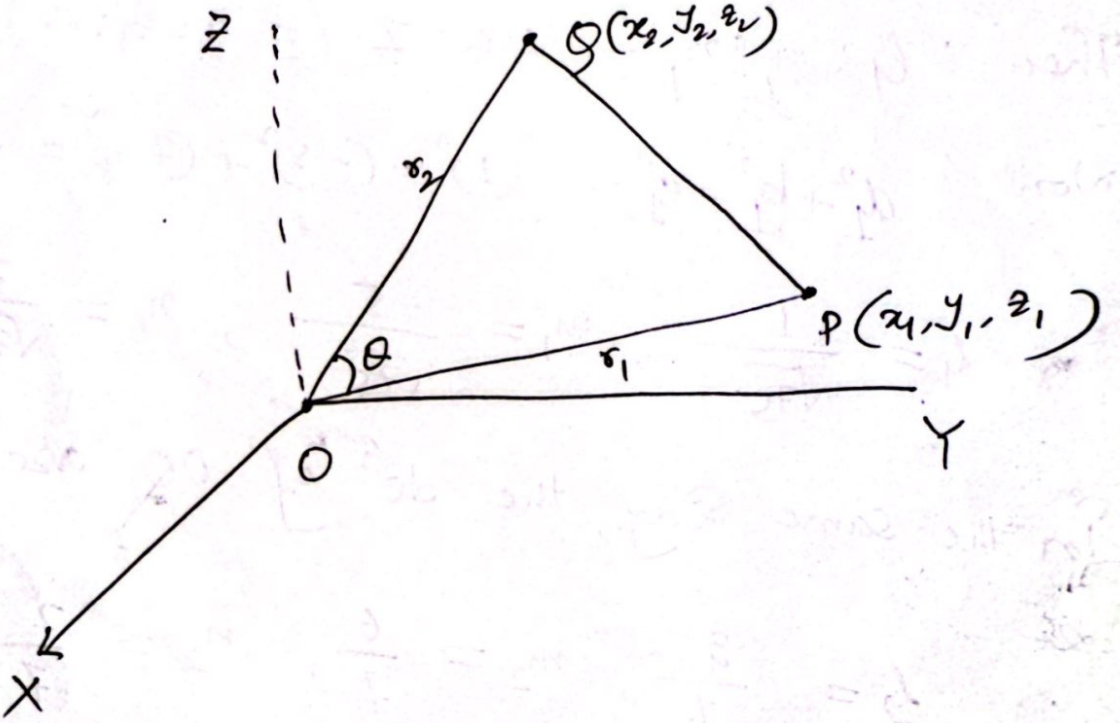
$$l = \frac{-4}{\sqrt{218}}, \quad m = \frac{11}{\sqrt{218}}, \quad n = \frac{-9}{\sqrt{218}}$$

Scanned with CamScanner

Angle bet^n two straight lines:

Th^m: If l_1, m_1, n_1 and l_2, m_2, n_2 be the d.c's of any two lines and θ be the angle between them then prove that $\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$.

Proof:



Let OP and OQ be two given straight lines having direction cosines l_1, m_1, n_1 and l_2, m_2, n_2 respectively. The angle between them be θ. Let $OP = r_1$, $OQ = r_2$ and the coordinates of P and Q be (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively. Then

$$\left.\begin{aligned} x_1 &= l_1 r_1, \quad y_1 = m_1 r_1, \quad z_1 = n_1 r_1 \\ x_2 &= l_2 r_2, \quad y_2 = m_2 r_2, \quad z_2 = n_2 r_2 \end{aligned}\right\} \text{—①}$$

Scanned with CamScanner

Then from trigonometry, we have

$$PQ^2 = OP^2 + OQ^2 - 2.OP.OQ\cos\theta \quad \text{(using cosine rule)}$$

$$\Rightarrow 2OP.OQ.\cos\theta = OP^2 + OQ^2 - PQ^2$$

$$\Rightarrow 2r_1r_2\cos\theta = r_1^2 + r_2^2 - \left[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2\right]$$

$$\Rightarrow 2r_1r_2\cos\theta = r_1^2 + r_2^2 - \left[x_2^2 + y_2^2 + z_2^2 + x_1^2 + y_1^2 + z_1^2 - 2(x_1x_2 + y_1y_2 + z_1z_2)\right]$$

$$\Rightarrow 2r_1r_2\cos\theta = r_1^2 + r_2^2 - r_2^2 - r_1^2 + 2(x_1x_2 + y_1y_2 + z_1z_2)$$

$$\Rightarrow r_1r_2\cos\theta = x_1x_2 + y_1y_2 + z_1z_2 \quad \text{---}$$

$$\Rightarrow r_1r_2\cos\theta = l_1r_1\, l_2r_2 + m_1r_1 \cdot m_2r_2 + n_1r_1\, n_2r_2 \quad \text{---}$$

$$\Rightarrow r_1r_2\cos\theta = r_1r_2(l_1l_2 + m_1m_2 + n_1n_2)$$

$$\Rightarrow \boxed{\cos\theta = l_1l_2 + m_1m_2 + n_1n_2} \quad \text{---} \quad (2)$$

Corollary-(1):

$$\sin\theta = \pm\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}$$

Soln: We know that

$$\cos\theta = l_1l_2 + m_1m_2 + n_1n_2 \quad \text{---}$$

$$\Rightarrow \cos^2\theta = (l_1l_2 + m_1m_2 + n_1n_2)^2$$

Scanned with CamScanner

$$\Rightarrow 1 - \sin^2\theta = (l_1 l_2 + m_1 m_2 + n_1 n_2)^2$$

$$\Rightarrow \sin^2\theta = 1 - (l_1 l_2 + m_1 m_2 + n_1 n_2)^2$$

$$\Rightarrow \sin^2\theta = 1.1 - (l_1 l_2 + m_1 m_2 + n_1 n_2)^2$$

$$= (l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1 l_2 + m_1 m_2 +$$

$$= l_1^2 l_2^2 + l_2^2 m_1^2 + l_2^2 n_1^2 + l_1^2 m_2^2 + m_1^2 m_2^2 + m_2^2$$
$$+ l_1^2 n_2^2 + m_1^2 n_2^2 + n_1^2 n_2^2 - l_1^2 l_2^2 - m_1^2 m_2^2 - n$$
$$- 2 l_1 l_2 m_1 m_2 - 2 m_1 m_2 n_1 n_2 - 2 l_1 l_2 n_1$$

$$= m_1^2 n_2^2 - 2 m_1 n_2 m_2 n_1 + m_2^2 n_1^2$$
$$+ n_1^2 l_2^2 - 2 n_1 l_2 n_2 l_1 + n_2^2 l_1^2$$
$$+ l_1^2 m_2^2 - 2 l_1 m_2 l_2 m_1 + l_2^2 m_1^2$$

$$\Rightarrow \sin^2\theta = (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2$$

$$\Rightarrow \boxed{\sin\theta = \pm\sqrt{(m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2 m_1)^2}}$$

— (3)

Again,

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

Scanned with CamScanner

$$\Rightarrow \tan\theta = \frac{\pm\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}}{l_1l_2 + m_1m_2 + n_1n_2} \quad \text{---- (4)}$$

Corollary (2):

Corresponding formula when direction ratios of the lines are given:

Let (a_1, b_1, c_1) and (a_2, b_2, c_2) be the d.r.s of the given lines OP and OQ, respectively, then their actual d.c.s are given by

$$l_1 = \pm \frac{a_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}}$$

$$\frac{l_1}{a_1} = \frac{m_1}{b_1} = \frac{n_1}{c_1} = \frac{\sqrt{l_1^2 + m_1^2 + n_1^2}}{\sqrt{a_1^2 + b_1^2 + c_1^2}}$$

$$\Rightarrow \frac{l_1}{a_1} = \frac{\pm 1}{\sqrt{a_1^2 + b_1^2 + c_1^2}}$$

$$m_1 = \pm \frac{b_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} \quad \& \quad n_1 = \pm \frac{c_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}}$$

$$\boxed{\frac{l_1}{a_1} = \frac{m_1}{b_1} = \frac{n_1}{c_1}}$$

$$\& \quad l_2 = \pm \frac{a_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$m_2 = \pm \frac{b_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}} \quad \& \quad n_2 = \pm \frac{c_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Scanned with CamScanner

Thus,

$$\cos\theta = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$\therefore \cos\theta = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\Sigma a_1^2} \cdot \sqrt{\Sigma a_2^2}} \quad \text{—— (5)}$$

and,

$$\sin\theta = \pm \frac{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - }}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

—— (6)

Corollary – (3) :

(Condition of perpendicularity) :

If the two lines OP & OQ are perpendicular to each other, then $\theta = 90^\circ$ or $\cos\theta = 0$.

Therefore, equ^n (2) becomes

$$l_1l_2 + m_1m_2 + n_1n_2 = 0$$

or, equ^n (5) becomes

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

} Required conditions

Department of Applied Mathematics
University of Dhaka
Dhaka-1000, Bangladesh

Corollary - (4):

(Condition of parallelism):

If the two lines are parallel to each other, then

$$\theta = 0^\circ \quad \text{or} \quad \cos\theta = 1, \quad \text{or} \quad \sin\theta = 0$$

Therefore, equ^n (3) reduces to

$$\sqrt{(m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2 m_1)^2} = 0$$

$$\Rightarrow (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2 m_1)^2 = 0$$

If the sum of squares added is zero, then their individuals squares should be zero

$$m_1 n_2 - m_2 n_1 = 0, \quad n_1 l_2 - n_2 l_1 = 0, \quad l_1 m_2 - l_2 m_1 = 0$$

$$\Rightarrow m_1 n_2 = m_2 n_1, \quad n_1 l_2 = n_2 l_1, \quad l_1 m_2 = l_2 m_1$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{n_1}{n_2}, \quad \frac{n_1}{n_2} = \frac{l_1}{l_2}, \quad \frac{l_1}{l_2} = \frac{m_1}{m_2}$$

Thus,

$$\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = \frac{\sqrt{l_1^2 + m_1^2 + n_1^2}}{\sqrt{l_2^2 + m_2^2 + n_2^2}} = \frac{1}{1}$$

$$\Rightarrow \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = 1$$

Scanned with CamScanner

whence,

$$l_1 = l_2, \quad m_1 = m_2, \quad n_1 = n_2 \qquad \text{— (7)}$$

which shows that when the lines are parallel their respective direction cosines are equal.

Similarly, when $\theta = 0^\circ$, equ^n (6) reduces to

$$(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2 = 0$$

$$\therefore \quad b_1c_2 - b_2c_1 = 0 \quad \Rightarrow \quad \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$c_1a_2 - c_2a_1 = 0 \quad \Rightarrow \quad \frac{c_1}{c_2} = \frac{a_1}{a_2}$$

$$a_1b_2 - a_2b_1 = 0 \quad \Rightarrow \quad \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\therefore \quad \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Thus when the two lines are parallel ($\theta = 0^\circ$) then the d.r^s of the two lines are proportional.

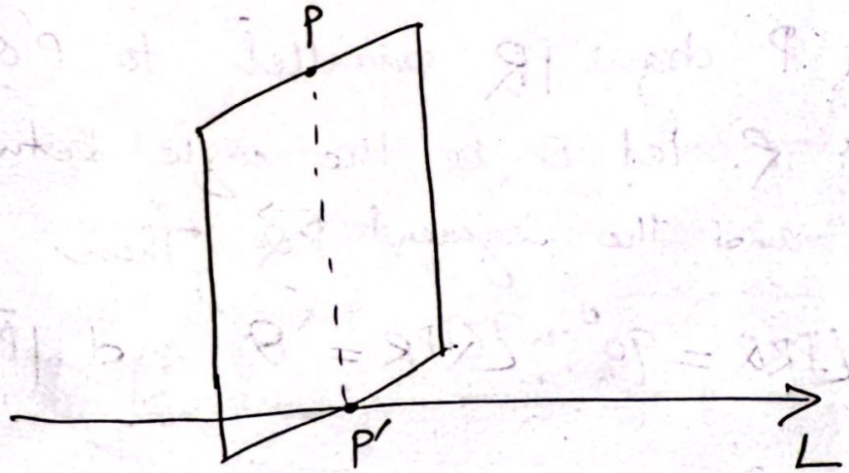
Lagrange's Identity: (H/W)

$$(a_1^2 + b_1^2 + c_1^2)(a_2^2 + b_2^2 + c_2^2) - (a_1a_2 + b_1b_2 + c_1c_2)^2$$
$$= (b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2$$

Projections

Orthogonal Projection.

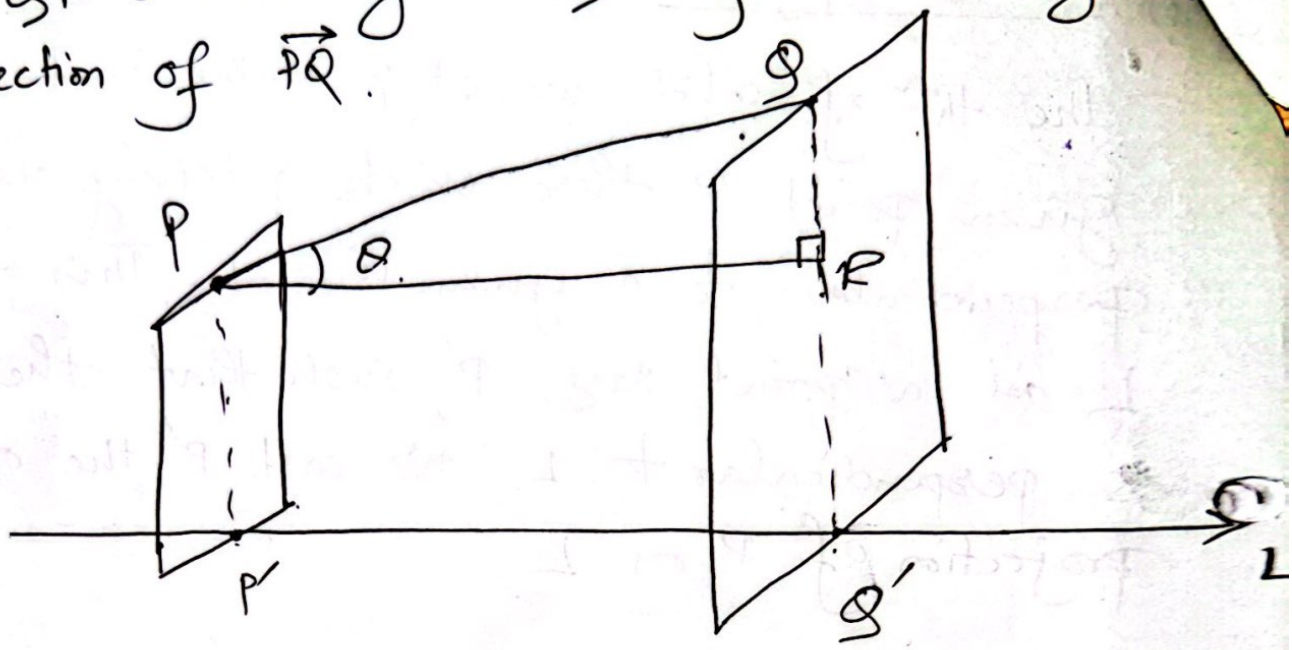
The th^{m} of solid geometry states that through a given point P there exists precisely one plane perpendicular to a given line L. This plane intersects L at a point, say, P′, such that the line PP′ is perpendicular to L. We call P′ the orthogonal projection of P on L.



Let P and Q be any two points. The line segment denoted by $\overrightarrow{PQ}$, which joins P to Q and which is directed from P to Q. Let the orthogonal projections of P and Q on a directed line L be P′ and Q′, respectively. Then we define the orthogonal projection of $\overrightarrow{PQ}$ on L to be signed length

Scanned with CamScanner

$\overrightarrow{P'Q'}$. It follows that the orthogonal projection of $\overrightarrow{QP}$ is the negative $\overrightarrow{Q'P'}$ of the orthogonal projection of $\overrightarrow{PQ}$.



Through P draw PR parallel to $P'Q'$ meeting QQ' in R. Let θ be the angle between the directed line L and the segment $\overrightarrow{PQ}$. Then

$\angle PRQ = 90^\circ$, $\angle QPR = \theta$ and $|\overrightarrow{PR}| = |\overrightarrow{P'Q'}|$

From the figure,

$$PR = PQ\cos\theta$$

or, $$\overrightarrow{P'Q'} = |\overrightarrow{PQ}|\cos\theta$$

$|\overrightarrow{PQ}|$ = length of the line segment $\overrightarrow{PQ}$.

Thus if θ be the angle between $\overrightarrow{PQ}$ and a directed line L, the projection of $\overrightarrow{PQ}$ on L is equal to $|\overrightarrow{PQ}|\cos\theta$

Scanned with CamScanner

Corollary (1):

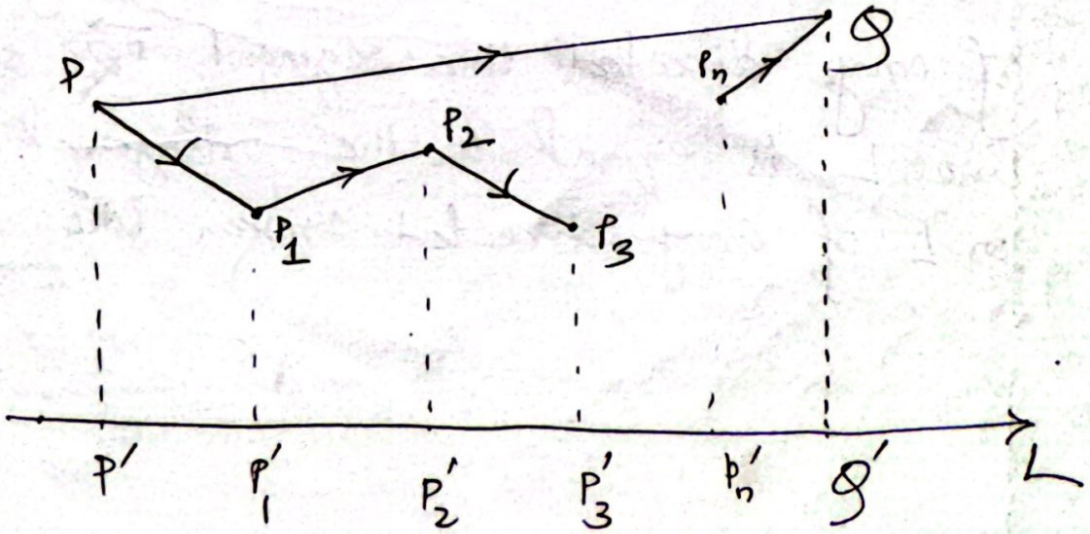
If the l.segment $\vec{PQ}$ is parallel to L, then $\theta = 0°$ or $\cos\theta = 1$, and the orthogonal projection of $\vec{PQ}$ on L is $|\vec{PQ}|$ (i.e., equal to its own length).

Corollary (2):

If the line segment $\vec{PQ}$ is perpendicular to the directed line L, then $\theta = 90°$ or $\cos\theta = 0$, and hence the projection of $\vec{PQ}$ on L is zero.

Thm:

A directed broken line joining two points P and Q is the geometrical configuration consisting of the directed line segments $\vec{PP_1}$, $\vec{P_1P_2}$, $\vec{P_2P_3}$, ..., $\vec{P_nQ}$ for any finite number of n-points $P_1, P_2, P_3, \ldots, P_n$.



Scanned with CamScanner

Let $P, P_1, P_2, P_3, \cdots, P_n, Q$ be the orthogonal projections of $P, P_1, P_2, P_3, \cdots, P_n, Q$, respectively, a directed line L. Then by definition,

projection of $\overrightarrow{PQ}$ upon $L = \overrightarrow{P'Q'}$

" " $\overrightarrow{PP_1}$ " $L = \overrightarrow{P'P_1'}$

" " $\overrightarrow{P_1P_2}$ " $L = \overrightarrow{P_1'P_2'}$

" " $\overrightarrow{P_2P_3}$ " $L = \overrightarrow{P_2'P_3'}$

" " $\overrightarrow{P_nQ}$ " $L = \overrightarrow{P_n'Q'}$

But

$$\overrightarrow{P'P_1'} + \overrightarrow{P_1'P_2'} + \overrightarrow{P_2'P_3'} + \cdots + \overrightarrow{P_n'Q'} = \overrightarrow{P'Q'}$$

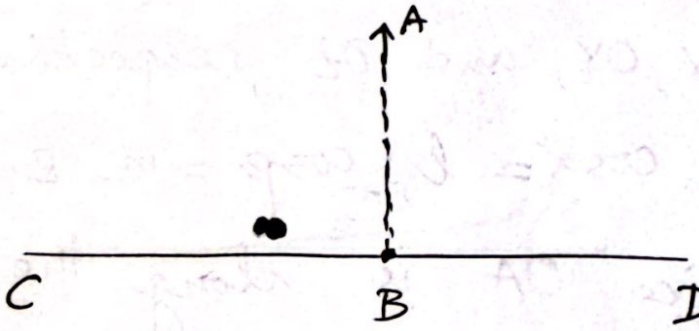
Thus we establish that the orthogonal projection of any directed line segment $\overrightarrow{PQ}$ on a directed line L is equal to the orthogonal projection on L of any directed broken line from P to Q.

Scanned with CamScanner

Projection of a point on a straight line:

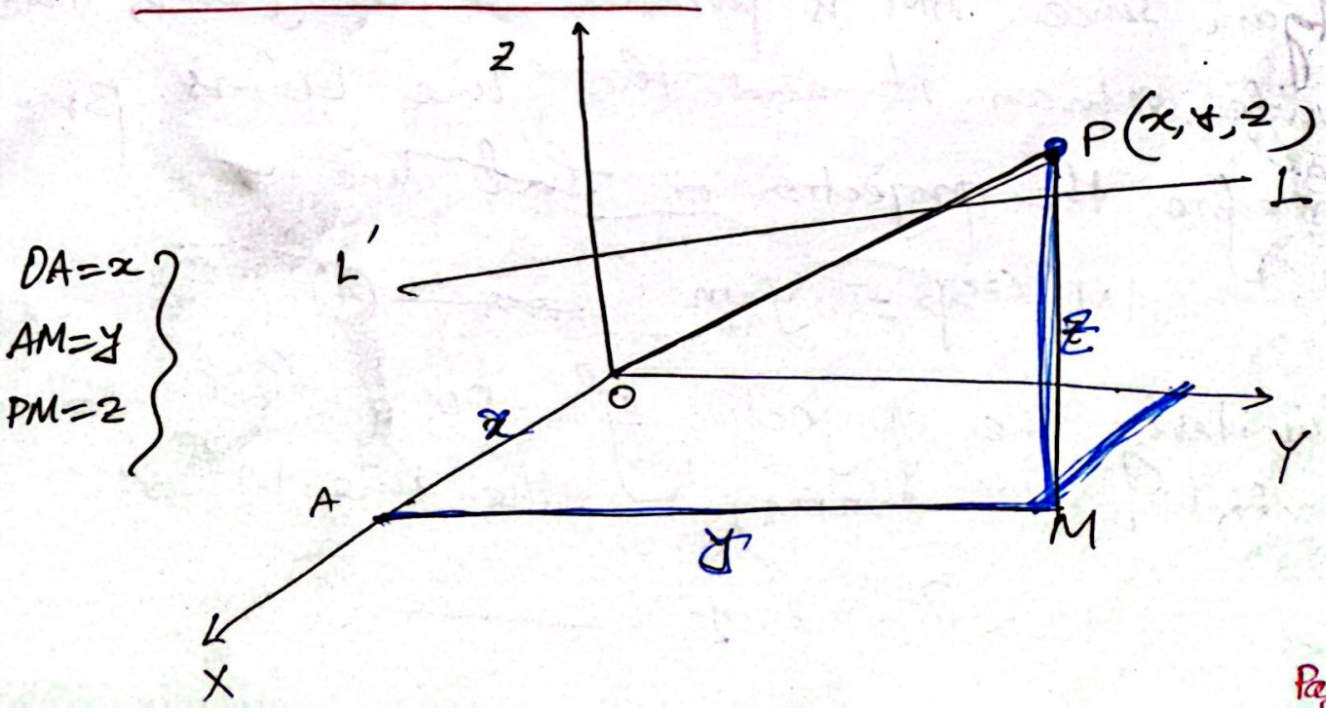
The projection of a point A on a given line CD is the point B which is the foot of the perpendicular from A on CD.

This is the point in which the plane through A and perpendicular to CD meets it.



29/12/2022

Projection of the joining of any point with the origin on a line :



Scanned with CamScanner

Let $P(x, y, z)$ be a point and LL' a line whos direction cosines are l, m, n.

Drop PM perpendicular on the XOY plane. Dra AM parallel to OY meeting OX at A.

Then $OA = x$; $AM = y$; $PM = z$; ——— (1)

Let the line LL' makes angles α, β and γ wi OX, OY, and OZ, respectively so that-

$\cos\alpha = l$, $\cos\beta = m$. & $\cos\gamma = n$ ——— (2)

Since OA is along the x-axis, the angle between it and the line LL' is α, and he its projection on the line is

$OA.\cos\alpha = x.l$ ——— (3)

Again, since AM is parallel to the y-axis th angle between it and the line LL' is β. Therefore, the projection on that line is

$AM.\cos\beta = y.m$ ——— (4)

Similarly, the projection of PM (which is parallel to the z-axis) on the line LL' is

$PM.\cos\gamma = z.n$ ——— (5)

Now the projection of OP on any line is equal to the sum of the projections of OA, AM, MP on the same line

Thus from (3), (4), (5) it follows that the projection of OP on the line LL' = $lx + my + nz$.

Hence, the projection of the joining of any point $P(x, y, z)$ with the origin on a directed line whose direction cosines are l, m, n is

$$lx + my + nz \qquad \text{——————} (6)$$

* **Note :**

The direction cosines of the x-axis are $(1, 0, 0)$.

Therefore, the projection of OP on the x-axis is

$1 \cdot x + 0 \cdot y + 0 \cdot z = x$.

Similarly, the projection of OP on the y-axis is

$0 \cdot x + 1 \cdot y + 0 \cdot z = y$

and the projection of OP on the z-axis is

$0 \cdot x + 0 \cdot y + 1 \cdot z = z$.

Scanned with CamScanner

Corollary:

The projection of the line segment between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ on a line with d.c.s l, m, n is $l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$.

Let $P_1(x_1, y_1, z_1)$ & $P_2(x_2, y_2, z_2)$ be two points. Transfer the origin to P_1 keeping the direction of the axes unchanged, and (x', y', z') be the coordinates of P_2 referred to these new set of axes. Then

$$x' = x_2 - x_1, \quad y' = y_2 - y_1 \quad \& \quad z' = z_2 - z_1 \qquad \text{---(7)}$$

Hence, using (6) the projection of P_1P_2 on a line whose direction cosines are l, m, n is

$$lx' + my' + nz' = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1) \qquad \text{---(8)}$$

Thus, the projection of a line segment between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ on a line with d.c.s l, m, n is

$$l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

Scanned with CamScanner

Problem: (Incourse 2022)

Find the projection of the joining of two points $(-1,-1,3)$ & $(2,0,1)$ on the line through the points $(-7,5,3)$ and $(2,6,8)$.

Soln:

Let l, m, n be the d.c.s of the line through the points $(-7,5,3)$ and $(2,6,8)$. Then we have,

$$\frac{l}{-7-2} = \frac{m}{5-6} = \frac{n}{3-8}$$

$$\Rightarrow \frac{l}{9} = \frac{m}{1} = \frac{n}{5} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{(9)^2+(1)^2+(5)^2}} = \frac{1}{\sqrt{107}}$$

Thus, $l = \frac{9}{\sqrt{107}}$; $m = \frac{1}{\sqrt{107}}$ & $n = \frac{5}{\sqrt{107}}$; ——— (1)

Now, the projection of the joining of the two points $(-1,-1,3)$ & $(2,0,1)$ on the given line is

$$l(x_2-x_1) + m(y_2-y_1) + n(z_2-z_1)$$

$$= l\{2-(-1)\} + m\{0-(-1)\} + n\{1-(3)\}$$

$$= \frac{9}{\sqrt{107}}(2+1) + \frac{1}{\sqrt{107}}(1) + \frac{5}{\sqrt{107}}(1-3)$$

$$= \frac{1}{\sqrt{107}}[(9*3) + (1*1) + 5*(-2)] = \frac{18}{\sqrt{107}}$$

Scanned with CamScanner

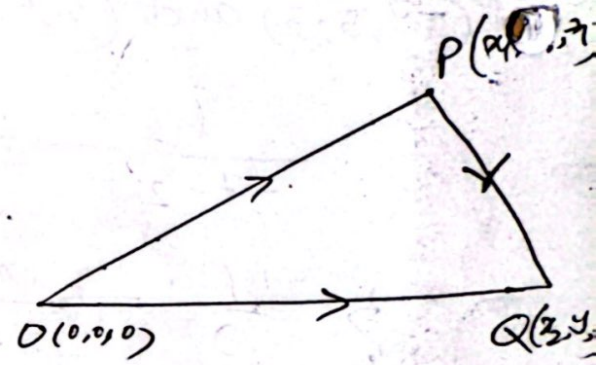
Problem: Show that the projection of a line joining the two points (x_1, y_1, z_1) & (x_2, y_2, z_2) on a straight line whose direction cosines are l, m, n is given by $l(x_2-x_1) + m(y_2-y_1) + n(z_2-z_1)$.

Soln: Let $O(0,0,0)$ be the origin and the given two points are $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$. Then

$\overrightarrow{OP} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ &

$\overrightarrow{OQ} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$

where, $\hat{i}, \hat{j}, \hat{k}$ are the unit directional vector along x, y, z axis, respectively.



Now $\overrightarrow{OP} + \overrightarrow{PQ} = \overrightarrow{OQ}$ [using triangle law of vector addition]

$\Rightarrow \overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$

$\therefore \overrightarrow{PQ} = (x_2-x_1)\hat{i} + (y_2-y_1)\hat{j} + (z_2-z_1)\hat{k}$.

Now, the unit vector along the line whose dc's are l, m, n is $l\hat{i} + m\hat{j} + n\hat{k}$

$\therefore$ Projection of PQ on the line whose dc's are l, m, n

$= \{(x_2-x_1)\hat{i} + (y_2-y_1)\hat{j} + (z_2-z_1)\hat{k}\} \cdot \{l\hat{i} + m\hat{j} + n\hat{k}\}$

$= l(x_2-x_1) + m(y_2-y_1) + n(z_2-z_1)$

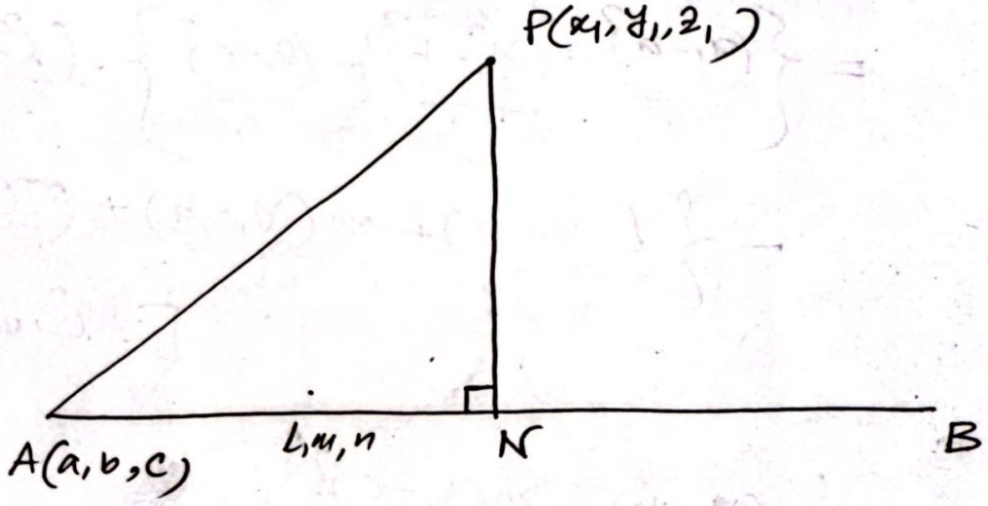
(Proved.)

Dr. Muntasir Alam
Assistant Professor
Department of Applied Mathematics
University of Dhaka
Dhaka-1000, Bangladesh

Th^m:

Find the perpendicular distance of a point $P(x_1, y_1, z_1)$ from a line through the point $A(a, b, c)$ and whose direction cosines are l, m, n.

Sol^n:



Let AB be a line through $A(a,b,c)$ and $P(x_1, y_1, z_1)$ be a point. Draw $PN \perp AB$ from P; join AP.

Now,

AN = Projection of AP upon AB.

$$= l(x_1 - a) + m(y_1 - b) + n(z_1 - c)$$

AP = Distance bet^n the two points A and P

$$= \sqrt{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2}$$

$$\Rightarrow AP^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2$$

From ΔAPN, using Pythagorous th^m we get

$$AP^2 = AN^2 + PN^2$$

Scanned with CamScanner

$$\Rightarrow PN^2 = AP^2 - AN^2$$

$$\Rightarrow PN^2 = \{(x_1-a)^2+(y_1-b)^2+(z_1-c)^2\} - \{l(x_1-a)+m(y_1-b)+n(z_1-c)\}^2$$

$$= \left\{\frac{(x_1-a)^2}{a_1^2}+\frac{(y_1-b)^2}{b_1^2}+\frac{(z_1-c)^2}{c_1^2}\right\}\cdot\left(\frac{l^2}{a_2^2}+\frac{m^2}{b_2^2}+\frac{n^2}{c_2^2}\right) - \left\{\frac{l(x_1-a)}{a_1a_2}+\frac{m(y_1-b)}{b_1b_2}+\frac{n(z_1-c)}{c_1c_2}\right\}^2$$

$$[\because l^2+m^2+n^2=1]$$

$$= \{n(y_1-b)-m(z_1-c)\}^2 + \{l(z_1-c)-n(x_1-a)\}^2 + \{m(x_1-a)-l(y_1-b)\}^2$$

[using Lagrange's Identity]

$$\therefore PN^2 = \begin{vmatrix} y_1-b & z_1-c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1-c & x_1-a \\ n & l \end{vmatrix}^2 + \begin{vmatrix} x_1-a & y_1-b \\ l & m \end{vmatrix}^2$$

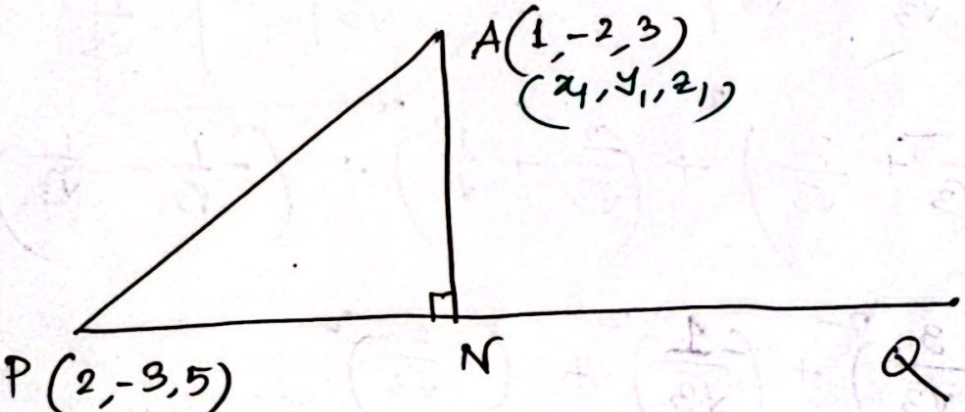
Problem: (Incourse 2022)

Find the distance of $A(1,-2,3)$ from the line PQ through $P(2,-3,5)$ which makes equal angles with the axes.

Soln:

Let l, m, n be the direction cosines of the line PQ.

Since, PQ makes equal angle α with x-axis, y-axis and z-axis, therefore $l = m = n$ (or, $\cos\alpha = \cos\beta = \cos\gamma$)



d.c.s: $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

But, $l^2 + m^2 + n^2 = 1$

$\Rightarrow 3l^2 = 1 \quad \Rightarrow l = \frac{1}{\sqrt{3}}$; Similarly, $m = \frac{1}{\sqrt{3}}$ & $n = \frac{1}{\sqrt{3}}$

$\therefore l = m = n = \frac{1}{\sqrt{3}}$.

If AN be the required perpendicular distance of A from the line PQ, then we have

$$AN^2 = \begin{vmatrix} y_1-b & z_1-c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1-c & x_1-a \\ n & l \end{vmatrix}^2 + \begin{vmatrix} x_1-a & y_1 \\ l & m \end{vmatrix}$$

$$= \begin{vmatrix} -2+3 & 3-5 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}^2 + \begin{vmatrix} 3-5 & 1-2 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}^2 + \begin{vmatrix} 1-2 & -2+3 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}^2$$

$$= \begin{vmatrix} 1 & -2 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}^2 + \begin{vmatrix} -2 & -1 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}^2 + \begin{vmatrix} -1 & 1 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}^2$$

$$= \left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}\right)^2 + \left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right)^2 + \left(-\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}}\right)^2$$

$$= \left(\frac{3}{\sqrt{3}}\right)^2 + \left(\frac{3}{\sqrt{3}}\right)^2 + \left(-\frac{2}{\sqrt{3}}\right)^2$$

$$= \frac{9}{3} + \frac{9}{3} + \frac{4}{3}$$

$$= \frac{14}{3}$$

$$\Rightarrow AN = \sqrt{\frac{14}{3}}$$

$$= \frac{\sqrt{14}}{3}$$

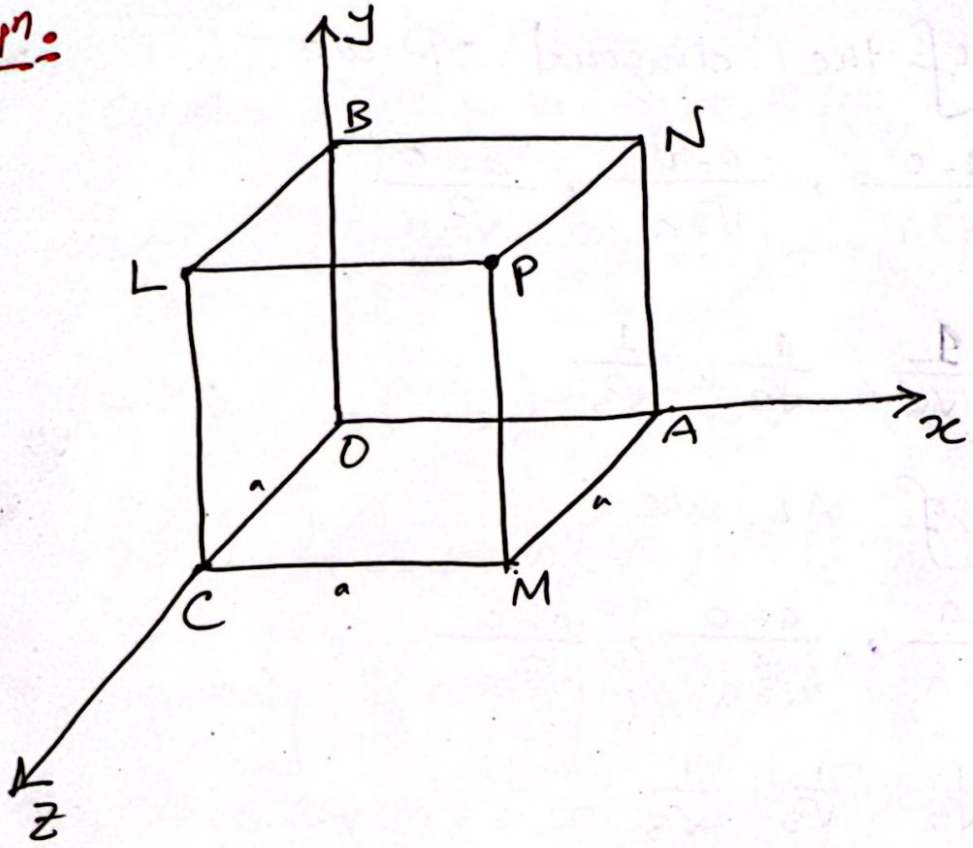
Ans)

Date: 9/9/22

Problem:

A line makes angle $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$.

Soln:



Let OCMANBLP be a cube whose edge is a.

The coordinates of the vertices are respectively

$O(0,0,0)$, $A(a,0,0)$, $B(0,a,0)$, $C(0,0,a)$, $M(a,0,a)$, $N(a,a,0)$, $L(0,a,a)$, $P(a,a,a)$;

Now,

$$OP = \sqrt{\{(0-a)^2 + (0-a)^2 + (0-a)^2\}}$$

$$= \sqrt{a^2 + a^2 + a^2}$$

$$= \sqrt{3}a$$

Scanned with CamScanner

Here, the four diagonals of the cube OP, AL, BM, CN are equal.

i.e., $OP = AL = BM = CN = \sqrt{3}a$

The dc's of the diagonal OP are:

$$\frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}$$

i.e., $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

The dc's of AL are:

$$\frac{0-a}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}$$

i.e, $-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$.

The dc's of BM are:

$$\frac{a-0}{\sqrt{3}a}, \frac{0-a}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}$$

i.e., $\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

The dc's of CN are:

$$\frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}, \frac{0-a}{\sqrt{3}a}$$

i.e., $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$

Scanned with CamScanner

Let l, m, n be the direction cosines of the line that makes angle $\alpha, \beta, \gamma, \delta$ with the four diagonals OP, AL, BM and CN, respectively,

Then,

$$\cos\alpha = l\cdot\left(\frac{1}{\sqrt{3}}\right) + m\cdot\left(\frac{1}{\sqrt{3}}\right) + n\cdot\left(\frac{1}{\sqrt{3}}\right) = \frac{l+m+n}{\sqrt{3}} \quad \text{---(1)}$$

$$\cos\beta = l\left(-\frac{1}{\sqrt{3}}\right) + m\left(\frac{1}{\sqrt{3}}\right) + n\left(\frac{1}{\sqrt{3}}\right) = \frac{-l+m+n}{\sqrt{3}} \quad \text{---(2)}$$

$$\cos\gamma = l\left(\frac{1}{\sqrt{3}}\right) + m\left(-\frac{1}{\sqrt{3}}\right) + n\left(\frac{1}{\sqrt{3}}\right) = \frac{l-m+n}{\sqrt{3}} \quad \text{---(3)}$$

$$\cos\delta = l\left(\frac{1}{\sqrt{3}}\right) + m\left(\frac{1}{\sqrt{3}}\right) + n\left(-\frac{1}{\sqrt{3}}\right) = \frac{l+m-n}{\sqrt{3}} \quad \text{---(4)}$$

Now, squaring & adding (1), (2), (3) & (4) we get.

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{1}{3}\left[(l+m+n)^2 + (-l+m+n)^2 + (l-m+n)^2 + (l+m-n)^2\right]$$

$$= \frac{1}{3}\left[l^2+m^2+n^2+2lm+2mn+2nl + l^2+m^2+n^2 - 2lm+2mn-2nl + l^2+m^2+n^2 - 2nl-2mn+2nl + l^2+m^2+n^2 + 2lm-2mn-2lm\right]$$

$$= \frac{4}{3}\left(l^2+m^2+n^2\right) = 4/3 \cdot (1) \quad [\because l^2+m^2+n^2 = 1]$$

$\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$

Proved

Scanned with CamScanner

Corollary: prove that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma + \sin^2\delta = 8/3$

We know, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$

$\Rightarrow 1 - \sin^2\alpha + 1 - \sin^2\beta + 1 - \sin^2\gamma + 1 - \sin^2\delta = 4/3$

$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma + \sin^2\delta = 4 - 4/3 =$

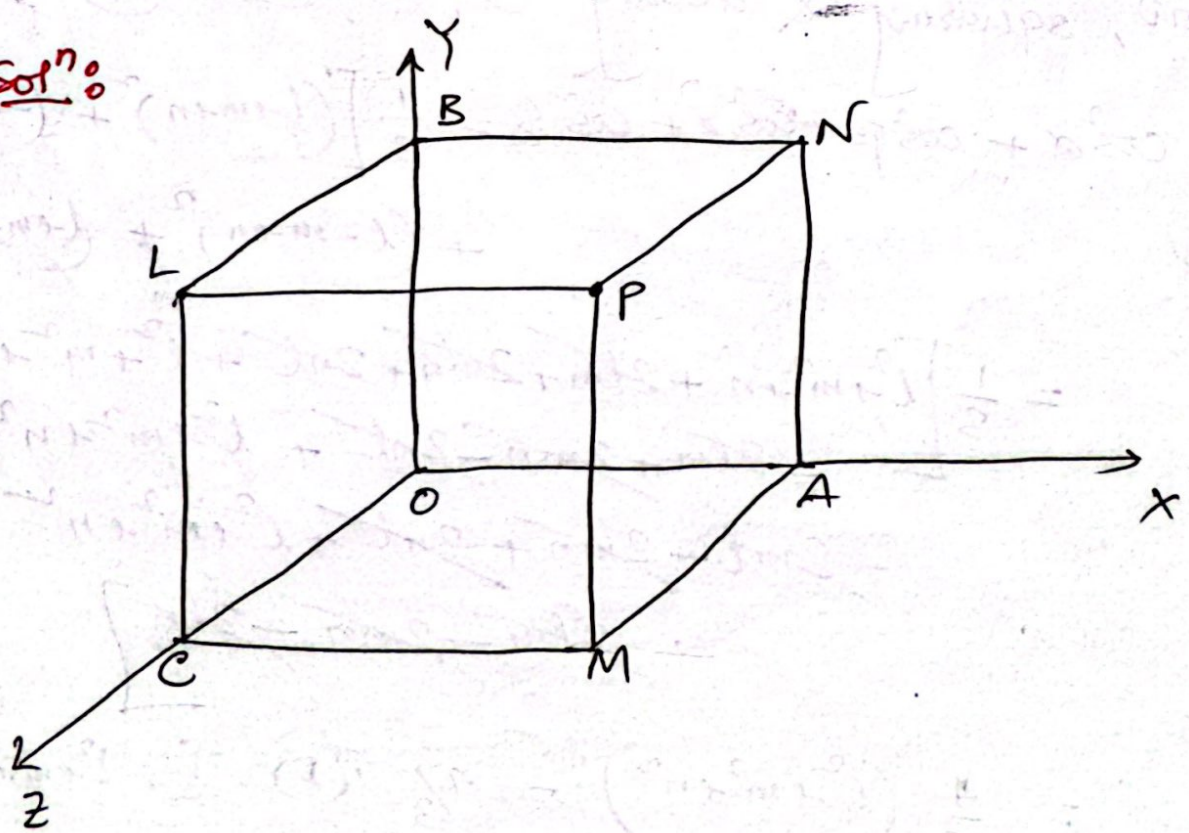
$\therefore \sin^2\alpha + \sin^2\beta + \sin^2\gamma + \sin^2\delta = 8/3$.

(Proved)

Problem:

Prove that the angle betⁿ two diagonals of a cube is $\cos^{-1}(1/3) \approx 70.53^0$

Solⁿ:



Let OCMANBLP be a cube whose edge is a. The coordinates of the vertices are respectively,

$O(0,0,0)$, $A(a,0,0)$, $B(0,a,0)$, $C(0,0,a)$, $M(a,0,a)$, $N(a,a,0)$, $L(0,a,a)$, $P(a,a,a)$;

The dr^s of the diagonal OP is:

$a-0, a-0, a-0$

i.e., a, a, a

The dr^s of the diagonal AL is

$0-a, a-0, a-0$

i.e., $-a, a, a$

Let, θ be the angle bet^n OP and AL

$$\therefore \cos\theta = \frac{a(-a) + a\cdot a + a\cdot a}{\sqrt{a^2+a^2+a^2}\cdot\sqrt{a^2+a^2+a^2}}$$

$$= \frac{-a^2+a^2+a^2}{\sqrt{3}a\cdot\sqrt{3}a}$$

$$\Rightarrow \cos\theta = \frac{a^2}{3a^2}$$

$$\Rightarrow \cos\theta = \frac{1}{3}$$

$$\therefore \theta = \cos^{-1}\left(\frac{1}{3}\right) \approx 70.53^0$$

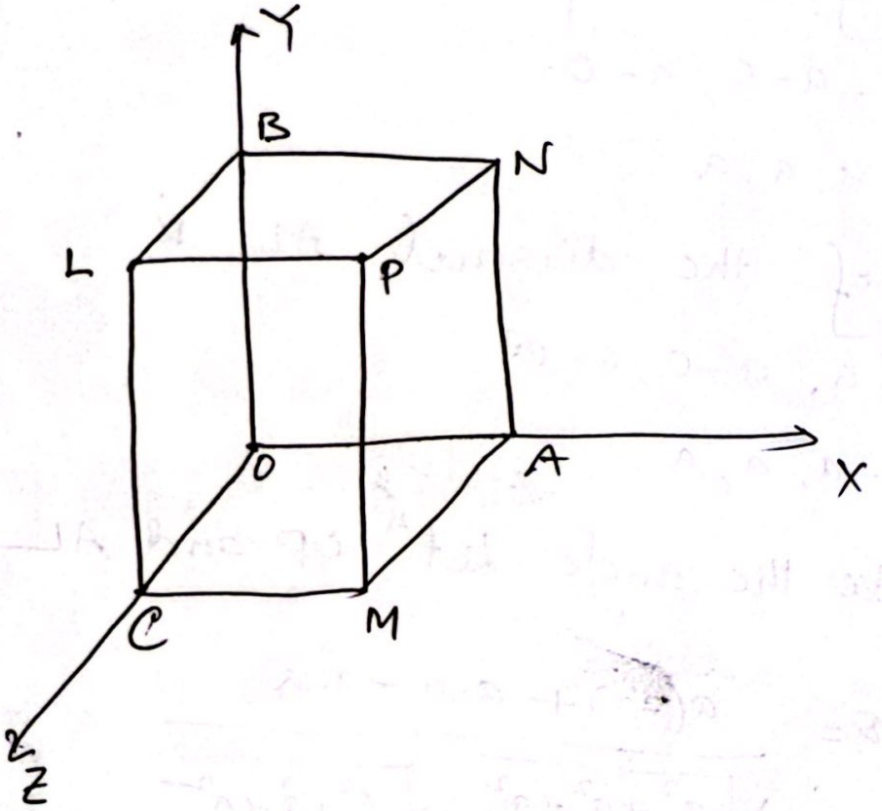
(Proved)

Scanned with CamScanner

Problem:

If the edges of a rectangular parallelopiped are a, b, c, show that the angle between the four diagonals are given by $\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}\right)$

Soln:



Take one of the vertices O of the parallelopiped as origin and three rectangular faces through it as the three rectangular coordinate planes.

Let $OA = a$, $OB = b$, $OC = c$.

The lines OP, AL, BM and CN are four diagonals of the parallelopiped. whose coordinates are

O (0,0,0), A (a,0,0), B (0,b,0), C (0,0,c), M (a,0,c)

Scanned with CamScanner

University of Dhaka
Dhaka-1000, Bangladesh

N(a, b, 0), L(0, b, c) and P(a, b, c)

DC's of OP: $\frac{a-0}{\sum a^2}, \frac{b-0}{\sum a^2}, \frac{c-0}{\sum a^2}$

i.e., $\frac{a}{\sum a^2}, \frac{b}{\sum a^2}, \frac{c}{\sum a^2}$

DC's of AL: $\frac{0-a}{\sum a^2}, \frac{b-0}{\sum a^2}, \frac{c-0}{\sum a^2}$

i.e., $\frac{-a}{\sum a^2}, \frac{b}{\sum a^2}, \frac{c}{\sum a^2}$

DC's of BM: $\frac{a-0}{\sum a^2}, \frac{0-b}{\sum a^2}, \frac{c-0}{\sum a^2}$

i.e., $\frac{a}{\sum a^2}, \frac{-b}{\sum a^2}, \frac{c}{\sum a^2}$

DC's of CN: $\frac{a-0}{\sum a^2}, \frac{b-0}{\sum a^2}, \frac{0-c}{\sum a^2}$

i.e., $\frac{a}{\sum a^2}, \frac{b}{\sum a^2}, \frac{-c}{\sum a^2}$

Let the angle betⁿ OP & AL is θ

$$\therefore \cos\theta = \frac{a(-a) + b(b) + c.(c)}{\sqrt{\sum a^2}.\sqrt{\sum a^2}}$$

$$\Rightarrow \cos\theta = \frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2}$$

Scanned with CamScanner

$$\therefore \theta = \cos^{-1}\left(\frac{-a^2+b^2+c^2}{a^2+b^2+c^2}\right).$$

Similarly, the angle between any one of the six pairs of diagonals can be found. ans

$${}^4C_2 = 6$$

Hence, the angles, in general, betⁿ the six pairs are given by,

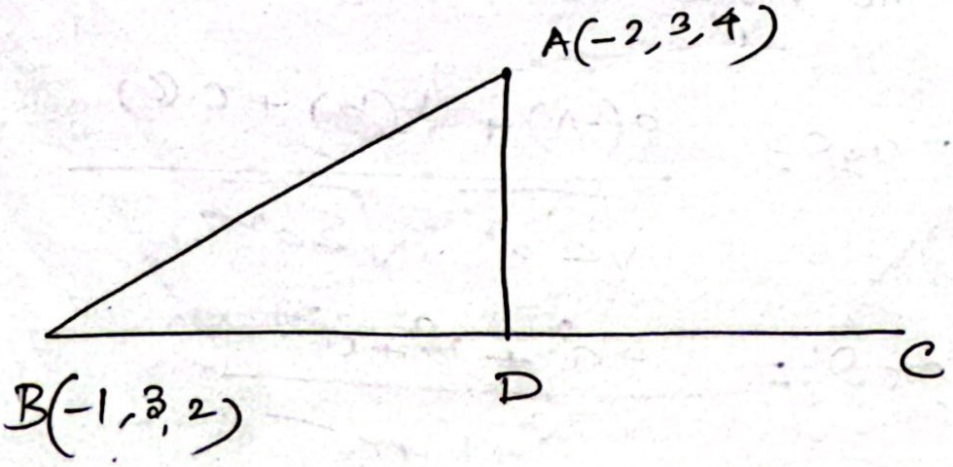
$$\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}\right)$$

(Proved)

Problem:

Find the distance of $(-2,3,4)$ from the line through the point $(-1,3,2)$ where the dc's are proportional to $12, 3, -4$.

Solⁿ:



Let BC be the line passing through the point B(-1,3,2) whose dc's are proportional to 12, 3, -4

Let, l, m, n be the dc's of BC

$$\frac{l}{12} = \frac{m}{3} = \frac{n}{-4} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{(12)^2+(3)^2+(-4)^2}} = \frac{1}{\sqrt{169}} = \frac{1}{13}$$

$$\therefore l = \frac{12}{13}, \quad m = \frac{3}{13}, \quad n = -\frac{4}{13}.$$

$$\therefore AB = \sqrt{(-2+1)^2 + (3-3)^2 + (4-2)^2} = \sqrt{1+0+4} = \sqrt{5}$$

Draw $AD \perp BC$ from the point A.

BD = The projection BA on BC

$$\Rightarrow BD = \frac{12}{13}(-2+1) + \frac{3}{13}(3-3) + \left(-\frac{4}{13}\right)(4-2)$$

$$= -\frac{12}{13} + 0 - \frac{8}{13}$$

$$= -\frac{20}{13}$$

From the right angled triangle ABD

$$AB^2 = AD^2 + BD^2$$

$$\Rightarrow AD = \sqrt{AB^2 - BD^2}$$

$$= \sqrt{(\sqrt{5})^2 - \left(-\frac{20}{13}\right)^2}$$

Scanned with CamScanner

$$\Rightarrow AD = \sqrt{5 - \frac{400}{169}}$$

$$= \frac{\sqrt{845 - 400}}{\sqrt{169}}$$

$$\therefore AD = \frac{\sqrt{445}}{13}$$

Example:

Show that the equation of a right circular cone whose vertex is at the origin, whose axes have dc's $\cos\alpha, \cos\beta, \cos\gamma$, and semivertical angle is θ is given by

$$(x\cos\beta - y\cos\alpha)^2 + (y\cos\gamma - z\cos\beta)^2 + (z\cos\alpha - x\cos\gamma)^2 = (x^2+y^2+z^2)\sin^2\theta.$$

Sol^n:

Let $P(x,y,z)$ be any point on the right circular cone. Take the circular section of the cone through the point P. Let C be the center of the circle

Then $PC \perp OC$

$$\therefore \sin\theta = \frac{PC}{OP}$$

$$\Rightarrow PC^2 = OP^2 \sin^2\theta \quad \text{——— (1)}$$

Scanned with CamScanner

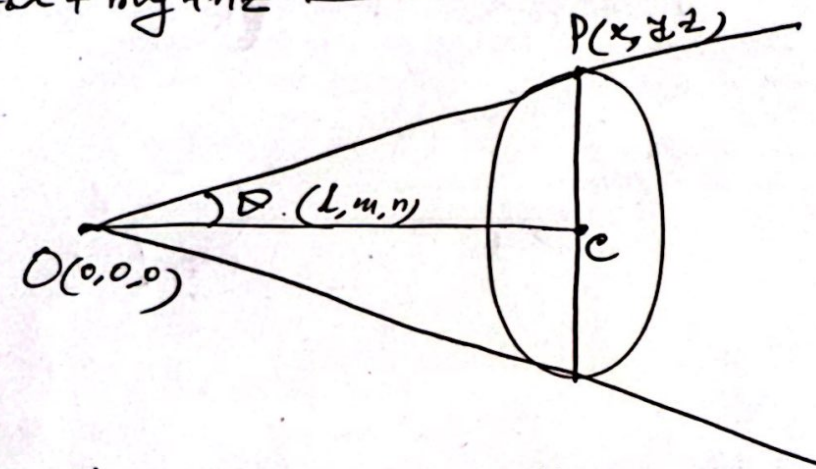
$$OP^2 = x^2 + y^2 + z^2 \longrightarrow (11)$$

OC = projection of OP on OC

$$= (x-0)\cos\alpha + (y-0)\cos\beta + (z-0)\cos\gamma$$

$$\Rightarrow OC = x\cos\alpha + y\cos\beta + z\cos\gamma \longrightarrow (17)$$

$$= lx + my + nz$$



Now, using Pythagorian th^m on the right-angled triangle OPC we have

$$OP^2 = OC^2 + PC^2$$

$$\Rightarrow PC^2 = OP^2 - OC^2$$

$$\Rightarrow PC^2 = (x^2 + y^2 + z^2) - (lx + my + nz)^2$$

$$\Rightarrow PC^2 = (x^2 + y^2 + z^2)(l^2 + m^2 + n^2) - (lx + my + nz)^2$$

$$= (yn - mz)^2 + (zl - nx)^2 + (xm - ly)^2$$

$$\Rightarrow PC^2 = (y\cos\gamma - z\cos\beta)^2 + (z\cos\alpha - x\cos\gamma)^2 + (x\cos\beta - y\cos\alpha)^2 \quad [\text{L. Identity}]$$

$$\Rightarrow OP^2\sin^2\theta = (x\cos\beta - y\cos\alpha)^2 + (y\cos\gamma - z\cos\beta)^2 + (z\cos\alpha - x\cos\gamma)^2$$

$$\Rightarrow PL^2 = a^2 + b^2 + c^2$$

$\therefore$ প্রথম সরলরেখার দিক-কোসাইন, $[l_1, m_1, n_1] = \left[-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

আবার (1) ও (4) বজ্রগুণন করে পাই, $\frac{l}{2} = \frac{m}{-1} = \frac{n}{2} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{4+1+4}} = \frac{1}{3}$

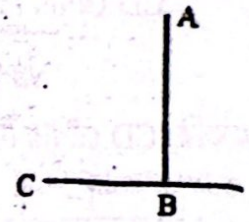
$\therefore$ দ্বিতীয় সরলরেখার দিক-কোসাইন, $[l_2, m_2, n_2] \equiv \left[\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right]$

এখন $l_1l_2 + m_1m_2 + n_1n_2 = -\frac{1}{3} \times \frac{2}{3} - \frac{2}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{2}{3} = 0$

সুতরাং সরলরেখাদ্বয় পরস্পর লম্ব। [Showed]

6.9 Projection of a point on a straight line
একটি বিন্দুর সরলরেখার উপর লম্ব অভিক্ষেপ

কোন বিন্দু A এর CD এর উপর লম্ব অভিক্ষেপ B বিন্দু যা A হতে CD এর উপর AB লম্বের পাদবিন্দু। A বিন্দুগামী তল CD সরলরেখা যে বিন্দুতে মিলিত হয় অর্থাৎ B বিন্দু A বিন্দুর লম্ব অভিক্ষেপ।

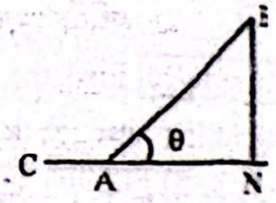


6.9.1 Projection of a line on a another straight line
একটি সরলরেখার উপর অপর একটি সরলরেখার লম্ব-অভিক্ষেপ

Theorem-3 : Prove that, the projection of the line AB, on the line CD, where the angle between two lines AB and CD is- [AB এবং CD এর মধ্যবর্তী কোণ θ CD রেখার উপর AB রেখার লম্ব অভিক্ষেপ হবে।]

$$AN = AB \cos\theta$$

Proof : Let CD be any line and AB be a line segment which intersect the line CD at the point A, where θ being the angle between two lines AB and CD. Draw perpendicular BN from B to CD. Then the projection of the line AB on the line CD is AN.



[মনে করি, CD যে কোন সরলরেখা এবং AB একটি রেখার খণ্ডাংশ, A তাদের ছে যেখানে AB এবং CD এর মধ্যবর্তী কোণ θ। B বিন্দু থেকে CD এর উপর BN লম্ব। তখন CD রেখার উপর AB রেখার লম্ব-অভিক্ষেপ AN]

From the right-angle triangle ABN, we get [ABN সমকোণী ত্রিভুজ থেকে, আমর

$$\cos\theta = \frac{AN}{AB}$$

$$\therefore AN = AB \cos\theta$$

Scanned with CamScanner

$\therefore$ প্রথম সরলরেখার দিক-কোসাইন, $[l_1, m_1, n_1] = \left[-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

আবার (1) ও (4) বজ্রগুণন করে পাই, $\frac{l}{2} = \frac{m}{-1} = \frac{n}{2} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{4+1+4}} = \frac{1}{3}$

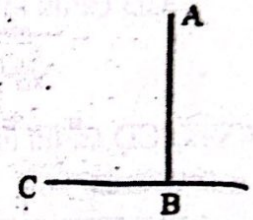
$\therefore$ দ্বিতীয় সরলরেখার দিক-কোসাইন, $[l_2, m_2, n_2] \equiv \left[\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right]$

এখন $l_1l_2 + m_1m_2 + n_1n_2 = -\frac{1}{3} \times \frac{2}{3} - \frac{2}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{2}{3} = 0$

সুতরাং সরলরেখাদ্বয় পরস্পর লম্ব। [Showed]

6.9 Projection of a point on a straight line
একটি বিন্দুর সরলরেখার উপর লম্ব অভিক্ষেপ

কোন বিন্দু A এর CD এর উপর লম্ব অভিক্ষেপ B বিন্দু যা A হতে CD এর উপর AB লম্বের পাদবিন্দু। A বিন্দুগামী তল CD সরলরেখা যে বিন্দুতে মিলিত হয় অর্থাৎ B বিন্দু A বিন্দুর লম্ব অভিক্ষেপ।

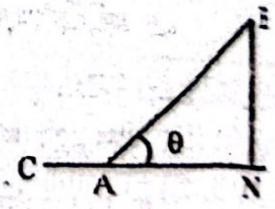


6.9.1 Projection of a line on a another straight line
একটি সরলরেখার উপর অপর একটি সরলরেখার লম্ব-অভিক্ষেপ

Theorem-3 : Prove that, the projection of the line AB, on the line CD, where the angle between two lines AB and CD is- [AB এবং CD এর মধ্যবর্তী কোণ θ CD রেখার উপর AB রেখার লম্ব অভিক্ষেপ হবে।]

$$AN = AB \cos\theta$$

Proof : Let CD be any line and AB be a line segment which intersect the line CD at the point A, where θ being the angle between two lines AB and CD. Draw perpendicular BN from B to CD. Then the projection of the line AB on the line CD is AN.



[মনে করি, CD যে কোন সরলরেখা এবং AB একটি রেখার খণ্ডাংশ, A তাদের ছে যেখানে AB এবং CD এর মধ্যবর্তী কোণ θ। B বিন্দু থেকে CD এর উপর BN লম্ব। তখন CD রেখার উপর AB রেখার লম্ব-অভিক্ষেপ AN]

From the right-angle triangle ABN, we get [ABN সমকোণী ত্রিভুজ থেকে, আমর

$$\cos\theta = \frac{AN}{AB}$$

$\therefore$ $AN = AB \cos\theta$

Scanned with CamScanner

Therefore, the projection of a given segment AB of a line on any line CD is AB cos θ, where θ be the angle between two lines AB and CD.

সুতরাং CD রেখার উপর যে কোন রেখার খণ্ডাংশ AB এর লম্ব-অভিক্ষেপ AB cos θ, যেখানে AB এবং CD এর মধ্যবর্তী কোণ θ] [Proved]

Example-9 Prove by the method of projection that, If A, B, C, D be four points whose co-ordinates are (6, – 6, 0), (– 1, – 7, 6), (3,– 4, 4) and (2, – 9, 2) respectively, then AB is perpendicular to CD.

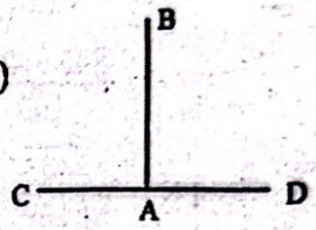
[অভিক্ষেপ পদ্ধতিতে প্রমাণ কর যে, A, B, C, D, চারটি বিন্দুর স্থানাঙ্ক যথাক্রমে (6, – 6, 0), (–1, –7, 6), (3, – 4, 4) এবং (2, – 9, 2) হলে AB রেখা CD রেখার উপর লম্ব।] *[DUH 2007]*

Solution প্রদত্ত বিন্দুগুলো,

$A(6, -6, 0), B(-1, -7, 6), C(3, -4, 4), D(2, -9, 2)$

CD রেখার দিক-অনুপাত, $3-2, -4+9, 4-2$

অর্থাৎ $1, 5, 2$



মনে করি, CD রেখার দিক-কোসাইন, l, m, n

$$\therefore l = \frac{1}{\sqrt{30}}, m = \frac{5}{\sqrt{30}}, n = \frac{2}{\sqrt{30}}$$

CD রেখার উপর AB এর অভিক্ষেপ $= (6+1)l + (-6+7)m + (0-6)n$

$$= 7.\frac{1}{\sqrt{30}} + 1.\frac{5}{\sqrt{30}} - 6.\frac{2}{\sqrt{30}}$$

$$= \frac{12}{\sqrt{30}} - \frac{12}{\sqrt{30}} = 0$$

যেহেতু অভিক্ষেপ শূন্য। অতএব, AB রেখা CD রেখার উপর লম্ব। [Proved]

Example-10 Find the perpendicular distance of the point (– 2, 3, 4) from the right line through the point (–1, 3, 2) whose direction ratio are 12, 3, – 4.

[–1, 3, 2) বিন্দুগামী যে সরলরেখার দিক-অনুপাত 12, 3, – 4 এর উপর (– 2, 3, 4) বিন্দু হতে লম্ব দূরত্ব নির্ণয় কর।] *[DUH 1986]*

Solution দেয়া আছে, (–1, 3, 2) বিন্দুগামী সরলরেখার দিক-অনুপাত 12, 3, – 4.

রেখাটির দিক-কোসাইন $\frac{12}{13}, \frac{3}{13}, \frac{-4}{13}$

এখন, APN একটি সমকোণী ত্রিভুজ।

এখন, $AP^2 = PN^2 + AN^2$

যেখানে, AN = AP এর অভিক্ষেপ

$$= (-2+1)\frac{12}{13} + (3-3)\frac{3}{13} + (4-2)\frac{(-4)}{13}$$

$$= -12 + 0 - 8 = \frac{-20}{13}$$

এবং $AP^2 = (-2+1)^2 + (3-3)^2 + (4-2)^2$

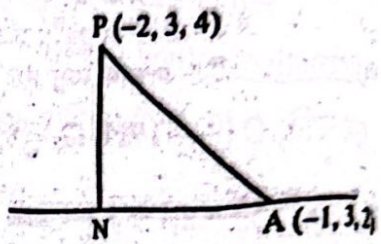
$= 1 + 0 + 4 = 5$

$\therefore PN^2 = AP^2 - AN^2$

$= 5 - \left(\frac{-20}{13}\right)^2$

$= 5 - \left(\frac{400}{169}\right)$

$= \frac{845 - 400}{169} = \frac{445}{169}$



$\therefore PN = \sqrt{\frac{445}{169}} = \frac{\sqrt{445}}{13}$

সুতরাং নির্ণেয় দূরত্ব $= \frac{\sqrt{445}}{13}$ (Ans.)

এখানে, AN = AP এর অভিক্ষেপ

$$= (-2+1)\frac{12}{13} + (3-3)\frac{3}{13} + (4-2)\frac{(-4)}{13}$$

$$= -12 + 0 - 8 = \frac{-20}{13}$$

এবং $AP^2 = (-2+1)^2 + (3-3)^2 + (4-2)^2$

$= 1 + 0 + 4 = 5$

Theorem-4 : Show that, the projection of a line joining the two points $(x_1, y_1, z$ and (x_2, y_2, z_2) on a straight line whose direction consines are l, m, n is [দেখাও (l, m, n দিক-কোসাইন বিশিষ্ট রেখার উপর (x_1, y_1, z_1) এবং (x_2, y_2, z_2) বিন্দুদ্বয়ের সংযে রেখার লম্ব-অভিক্ষেপ]

$$l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

Proof : Let O(0, 0, 0) be the origin and the given two points are $P(x_1, y_1, z_1)$ a $Q(x_2, y_2, z_2)$. Then

[মনে করি, O (0, 0, 0) মূলবিন্দু এবং $P(x_1, y_1, z_1)$ ও $Q(x_2, y_2, z_2)$ প্রদত্ত বিন্দুদ্বয়, তখন]

$$\vec{OP} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k} \text{ and } \vec{OQ} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

$$\therefore \vec{PQ} = \vec{OQ} - \vec{OP} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

Scanned with CamScanner

Scanned with CamScanner

ow the unit vector along the line whose direction consies are l, m, n [এখন l, m, n দিক-কোসাইন বিশিষ্ট রেখা বরাবর একক ভেক্টর] $= l\hat{i} + m\hat{j} + n\hat{k}$

. Projection of PQ on the line whose direction cosines are l, m, n [সুতরাং l, m, n দিক-কোসাইন বিশিষ্ট রেখার উপর PQ রেখার লম্ব-অভিক্ষেপ ।]

$$= \{(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}\} \cdot \{l\hat{i} + m\hat{j} + n\hat{k}\}$$
$$= l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1) \text{ [Showed]}$$

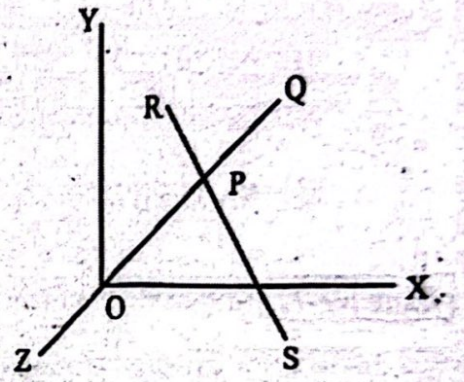
lternative method :

লে করি, O (0, 0, 0) মূলবিন্দু এবং প্রদত্ত বিন্দুদ্বয়, $P(x_1, y_1, z_1)$ এবং $Q(x_2, y_2, z_2)$ ।

$$\therefore \vec{OP} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$

এবং $\vec{OQ} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$

$$\therefore \vec{PQ} = \vec{OQ} - \vec{OP}$$
$$= l(x_2 - x_1)\hat{i} + m(y_2 - y_1)\hat{j} + n(z_2 - z_1)\hat{k}$$



খন l, m, n দিক-কোসাইন বিশিষ্ট RS রেখা বরাবর একক ভক্টর, $l\hat{i} + m\hat{j} + n\hat{k}$

তরাং RS রেখার উপর PQ রেখার লম্ব অভিক্ষেপ,

$$= \{(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}\} \cdot \{l\hat{i} + m\hat{j} + n\hat{k}\}$$
$$= (x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n \text{ [Showed]}$$

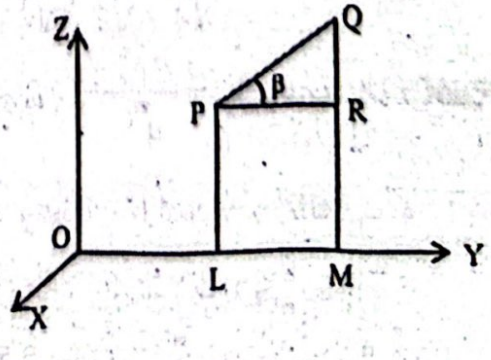
heorem-5 : If the co-ordinates of two points P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively then show that, direction ratios of PQ are $x_2 - x_1, y_2 - y_1, z_2 - z_1$ or $x_1 - x_2, y_1 - y_2, z_1 - z_2$. [যদি দুটি বিন্দু P ও Q এর স্থানাঙ্ক যথাক্রমে (x_1, y_1, z_1) এবং (x_2, y_2, z_2) হয় তবে দেখাও যে, PQ সরলরেখার দিক-অনুপাত $x_2 - x_1, y_2 - y_1, z_2 - z_1$ অথবা, $x_1 - x_2, y_1 - y_2, z_1 - z_2$]

roof : P ও Q বিন্দু হতে Y অক্ষের উপর PL ও QM এবং P বিন্দু হতে QM এর উপর PR ম অঙ্কন করি ।

$\therefore LM = PR = y_2 - y_1$

দি PQ সরলরেখা X, Y, Z অক্ষত্রয়ের সাথে থাক্রমে α, β, γ কোণ উৎপন্ন করে তবে $\angle QPR = \beta$

খন ΔPQR হতে $\cos\beta = \frac{PR}{PQ} \Rightarrow PQ = \frac{y_2 - y_1}{\cos\beta}$



নরূপভাবে Z অক্ষ ও X অক্ষের উপর লম্ব অংকন করে প্রমাণ করা যায়,

Scanned with CamScanner

$$PQ = \frac{z_2 - z_1}{\cos\gamma} \text{ এবং } PQ = \frac{x_2 - x_1}{\cos\beta}$$

সুতরাং $PQ = \frac{x_2 - x_1}{\cos\alpha} = \frac{y_2 - y_1}{\cos\beta} = \frac{z_2 - z_1}{\cos\gamma}$ ..... (1)

এখন PQ সরলরেখার দিক-কোসাইন l, m, n হলে,

$$l = \cos\alpha,\ m = \cos\beta,\ n = \cos\gamma$$

$$\Rightarrow \frac{x_2 - x_1}{l} = \frac{y_2 - y_1}{m} = \frac{z_2 - z_1}{n}$$

$$\Rightarrow \frac{l}{x_2 - x_1} = \frac{m}{y_2 - y_1} = \frac{n}{z_2 - z_1}$$

অথবা, $\frac{l}{x_1 - x_2} = \frac{m}{y_1 - y_2} = \frac{n}{z_1 - z_2}$

∴ সরলরেখার দিক-অনুপাত হলো,

$x_2 - x_1, y_2 - y_1, z_2 - z_1$ অথবা, $x_1 - x_2, y_1 - y_2, z_1 - x_2$ [Showed]

Example-11 অক্ষরেখাত্রয় একটি ঘনকের (cube) কর্ণের সাথে $\alpha, \beta, \gamma, \delta$ কোণ উৎপন্ন করে। প্রমাণ কর যে, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$.

[DU (7C) 2017; NUH 2000, 2002, 2007, 2019; RUH '88; DUH '87, '9

Solution মনে করি, সরলরেখাটির দিক-কোসাইন l, m, n এবং ঘনকের আটটি শীর্ষবিন্দু স্থানাঙ্ক O(0, 0, 0), P(a, a, a), A(a, 0, 0), B(0, a, 0), C(0, 0, a), L(0, a, a), M(a, 0, a) N(a, a, 0).

AL, BM, CN এবং OP ঘনকের কর্ণ। চারটি কর্ণের দিক-কোসাইনসমূহ যথাক্রমে $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

$-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}; \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}; \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$

চারটি কর্ণের সাথে $\alpha, \beta, \gamma, \delta$ কোণের আনত সরলরেখার দিক-কোসাইন l, m, n হলে,

$$\cos\alpha = \frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} = \frac{l + m + n}{\sqrt{3}}$$

অনুরূপভাবে, $\cos\beta = \frac{-l + m + n}{\sqrt{3}}$, $\cos\gamma = \frac{l - m + n}{\sqrt{3}}$, $\cos\delta = \frac{l + m - n}{\sqrt{3}}$

$$\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{1}{3}.4(l^2 + m^2 + n^2) = \frac{4}{3} \text{ [Proved]}$$

Scanned with CamScanner

Example-12 যদি l_1, m_1, n_1 ; l_2, m_2, n_2 ; l_3, m_3, n_3 তিনটি পরস্পর লম্ব সরলরেখার দিক-কোসাইন হয়, তবে দেখাও যে, $l_1 + l_2 + l_3$, $m_1 + m_2 + m_3$ এবং $n_1 + n_2 + n_3$ দিক-কোসাইন বিশিষ্ট একটি সরলরেখার সাথে এই সরলরেখাগুলো সমান কোণ উৎপন্ন করবে।

[RUH '86, '88]

Solution দেওয়া আছে, l_1, m_1, n_1; l_2, m_2, n_2 এবং l_3, m_3, n_3 দিক-কোসাইন বিশিষ্ট তিনটি সরলরেখা পরস্পর লম্ব।

$\therefore l_1l_2 + m_1n_1 + n_1n_2 = 0, l_2l_3 + m_2n_3 + n_2n_3 = 0$ ..... (1)

$l_1l_3 + m_1n_3 + n_1n_3 = 0$ ..... (2)

এবং $l_1^2 + m_1^2 + n_1^2 = 1, l_2^2 + m_2^2 + n_2^2 = 1, l_3^2 + m_3^2 + n_3^2 = 1$ ..... (3)

ধরি, l_1, l_2, l_3; m_1, m_2, m_3 এবং n_1, n_2, n_3 দিক-অনুপাত বিশিষ্ট সরলরেখাটি পরস্পর সরলরেখা তিনটির সাথে যথাক্রমে (α, β, γ) কোণ উৎপন্ন করে। প্রমাণ করতে হবে যে, $\alpha = \beta = \gamma$

$$\therefore \cos\alpha = \frac{l_1(l_1, l_2, l_3) + m_1(m_1, m_2, m_3) + n_1(n_1, n_2, n_3)}{\sqrt{l_1^2 + m_1^2 + n_1^2}\sqrt{(l_1, l_2, l_3)^2 + (m_1, m_2, m_3)^2 + (n_1, n_2, n_3)^2}}$$

$$= \frac{1 + 0 + 0}{\sqrt{1}\sqrt{1 + 1 + 1}} \quad [(1), (2) \text{ এবং } (3) \text{ এর সাহায্যে}]$$

$$= \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \cos^{-1}\frac{1}{\sqrt{3}}$$

$$\therefore \cos\beta = \frac{l_2(l_1, l_2, l_3) + m_1(m_1, m_2, m_3) + n_1(n_1, n_2, n_3)}{\sqrt{l_1^2 + m_1^2 + n_1^2}\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$= \frac{0 + 1 + 0}{\sqrt{1}\sqrt{1 + 1 + 1 + 2.0 + 2.0 + 2.0}} \quad [(1), (2) \text{ এবং } (3) \text{ এর সাহায্যে}]$$

$$= \frac{1}{\sqrt{3}}$$

$$\therefore \beta = \cos^{-1}\frac{1}{\sqrt{3}}$$

এবং,

$$\therefore \cos\gamma = \frac{l_3(l_1, l_2, l_3) + m_3(m_1, m_2, m_3) + n_3(n_1, n_2, n_3)}{\sqrt{l_1^2 + m_1^2 + n_1^2}\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

Scanned with CamScanner

$$= \frac{0+0+1}{\sqrt{1}\sqrt{1+1+1+2.0+2.0+2.0}} \quad [(1), (2) \text{ এবং } (3) \text{ এর সাহায্যে}]$$

$$= \frac{1}{\sqrt{3}}$$

$$\Rightarrow \cos\gamma = \frac{1}{\sqrt{3}}$$

$$\therefore \gamma = \cos^{-1}\frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \beta = \gamma$$

অর্থাৎ প্রত্যেক কোণ সমান হবে। [Showed]

Theorem-6 : If θ be the angle between two straight lines whose direction cosi are l_1, m_1, n_1 and l_2, m_2, n_2 then show that [l_1, m_1, n_1 এবং l_2, m_2, n_2 দিক-কোসা বিশিষ্ট সরলরেখাদ্বয়ের মধ্যবর্তী কোণ θ হলে দেখাও যে,]

(i) $\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$ *[NUH 2012, 20*

(ii) $\sin\theta = \pm\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}$ *[NUH 20*

(iii) $\tan\theta = \pm\dfrac{\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}}{l_1l_2 + m_1m_2 + n_1n_2}$

Also find the conditions such that the two straight lines are perpendicular a parallel to each other. [সরলরেখাদ্বয় পরস্পর লম্ব এবং সমান্তরাল হওয়ার শর্তও নি কর।] *[NUH '01, '02, '03, '09; NUH(NM) '*

Proof : Let, AB and CD be two given lines whose direction cosines are l_1, m_1, and l_2, m_2, n_2 respectively. Through the origin O draw lines OP_1 and OP_2 parall to the lines AB and CD respectively so that direction cosines of OP_1 and OP_2 a l_1, m_1, n_1 and l_2, m_2, n_2 respectively. [মনে করি, AB এবং CD প্রদত্ত দুটি সরলরে যাদের দিক-কোসাইন যথাক্রমে l_1, m_1, n_1 এবং l_2, m_2, n_2। AB এবং CD এর সমান্তর করে O মূলবিন্দুগামী OP_1 এবং OP_2 দুটি রেখা আঁকি। OP_1 এবং OP_2 রেখা প্রদত্ত AB এ CD রেখার সমান্তরাল হওয়ায় OP_1 এবং OP_2 রেখার দিক-কোসাইনও যথাক্রমে l_1, m_1, এবং l_2, m_2, n_2।]

Take a point $P_1(x_1, y_1, z_1)$ on OP_1 such that $OP_1 = r_1$. Since the direction cosines OP_1 are l_1, m_1, n_1; therefore the co-ordinates of P_1 may be written as [OP_1 রে উপর $P_1(x_1, y_1, z_1)$ বিন্দুটি এমনভাবে নিই, যেন $OP_1 = r_1$ হয়। যেহেতু OP_1 রেখার দি কোসাইন l_1, m_1, n_1, সুতরাং P_1 বিন্দুর স্থানাঙ্ক নিম্নরূপে লিখা যায়]

$$(l_1r_1, m_1r_1, n_1r_1)$$

$$\therefore x_1 = l_1r_1, y_1 = m_1r_1, z_1 = n_1r_1 \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

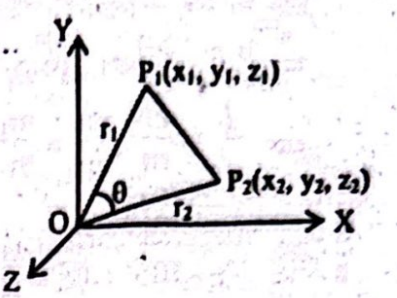
Scanned with CamScanner

ilarly, take a point $P_2(x_2, y_2, z_2)$ on OP_2 such the $OP_2 = r_2$. Since the direction ines of OP_2 are l_2, m_2, n_2; therefore the co-ordinates of P_2 may be written as

রূপভাবে, OP_2 রেখার উপর $P_2(x_2, y_2, z_2)$ বিন্দুটি এমনভাবে নিই, যেন $OP_2 = r_2$ হয়। তু OP_2 রেখার দিক-কোসাইন l_2, m_2, n_2; সুতরাং P_2 বিন্দুর স্থানাঙ্ক নিম্নরূপে লিখা যায়।]

(l_2r_2, m_2r_2, n_2r_2)

$\therefore x_2 = l_2r_2, y_2 = m_2r_2, z_2 = n_2r_2$ ..... (2)



θ be the angle between the two lines AB and then θ is the angle between OP_1 and OP_2. [মনে AB এবং CD রেখার মধ্যবর্তী কোণ θ, তাহলে এবং OP_2 রেখার মধ্যবর্তী কোণ ও θ]

projection of line OP_2 on the line OP_1 whose direction cosines are l_1, m_1, n_1 is

$l_1(x_2 - 0) + m_1(y_2 - 0) + n_1(z_2 - 0)$

$= l_1x_2 + m_1y_2 + n_1z_2$ ..... (3)

the projection of OP_2 on OP_1 is [কিন্তু OP_1 রেখার উপর OP_2 রেখার লম্ব অভিক্ষেপ]

$= OP_2 \cos\theta$ ..... (4)

From (3) and (4), we get [(3) নং এবং (4) নং থেকে, আমরা পাই]

$OP_2 \cos\theta = l_1x_2 + m_1y_2 + n_1z_2$

$\Rightarrow r_2 \cos\theta = l_1l_2r_2 + m_1m_2r_2 + n_1n_2r_2$ [using (2)]

$\therefore \cos\theta = l_1l_2 + m_1m_2 + n_1n_2$ [Showed]

$\cos^2\theta = (l_1 l_2 + m_1 m_2 + n_1 n_2)^2$

$\Rightarrow \sin^2\theta = 1 - (l_1 l_2 + m_1 m_2 + n_1 n_2)^2$

$= (l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1 l_2 + m_1 m_2 + n_1 n_2)^2$

$[\because l^2 + m^2 + n^2 = 1]$

$= l_1^2 m_2^2 + l_1^2 n_2^2 + m_1^2 l_2^2 + m_1^2 n_2^2 + n_1^2 l_2^2 + n_1^2 m_2^2 - 2l_1l_2 m_1m_2 - 2m_1m_2n_1n_2 - 2n_1n_2 l_1 l_2$

$= (l_1 m_2 - l_2 m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1 l_2 - n_2 l_1)^2$

$\therefore \sin\theta = \pm\sqrt{(l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2 l_1)^2}$ [Showed]

We know,

$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

$\Rightarrow \tan\theta = \pm\dfrac{\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1m_2 - l_2m_1)^2}}{l_1 l_2 + m_1m_2 + n_1n_2}$ [Showed]

ম হওয়ার শর্ত : যদি রেখাদ্বয় পরস্পর লম্ব হয়, তবে $\theta = 90°$ হবে এক্ষেত্রে,

(3) $\Rightarrow l_1 l_2 + m_1m_2 + n_1n_2 = \cos 90° = 0$ (Ans.)

Scanned with CamScanner

Scanned with CamScanner

• সমান্তরাল হওয়ার শর্ত : যদি সরলরেখাদ্বয় পরস্পর সমান্তরাল হয়, তবে $\theta = 0$ হবে এবং

$(4) \Rightarrow \sin^2 0 = (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1 l_2 - n_2 l_1)^2$

$\Rightarrow (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1 l_2 - n_2 l_1)^2 = 0$

$\because \Rightarrow (l_1m_1 - l_2m_1)^2 = 0, (m_1n_2 - m_2n_1)^2 = 0, (n_1 l_2 - n_2 l_1)^2 = 0$

$\Rightarrow l_1m_2 = l_2m_1, m_1n_2 = m_2n_1, n_1 l_2 = n_2 l_1$

$\Rightarrow \frac{l_1}{l_2} = \frac{m_1}{m_2}, \frac{m_1}{m_2} = \frac{n_1}{n_2}, \frac{n_1}{n_2} = \frac{l_1}{l_2}$

$\Rightarrow \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$ (Ans.)

Example-13 If the co-ordinates of P and Q be (2, 3, 4) and (1, 1, –1) respectively and O is the origin then find the angle between the line OP and OQ.
[P ও Q বিন্দুদ্বয়ের স্থানাঙ্ক (2, 3, 4) ও (1, 1, – 1) এবং O মূলবিন্দু হলে OP ও OQ সরলরেখাদ্বয়ের মধ্যবর্তী কোণ নির্ণয় কর।] *[RUH '80,*

Solution এখানে, OP সরলরেখার দিক-অনুপাত $2-0, 3-0, 4-0$ বা, 2, 3, 4

এবং OQ সরলরেখার দিক-অনুপাত 1, 1, – 1.

মনে করি, OP এবং OQ সরলরেখাদ্বয়ের মধ্যবর্তী কোণ θ.

$$\therefore \cos\theta = \frac{2.1 + 3.1 + 4.(-1)}{\sqrt{2^2+3^2+4^2}\sqrt{1^2+1^2+1^2}} = \frac{1}{\sqrt{29}.\sqrt{3}} = \frac{1}{\sqrt{87}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{1}{\sqrt{87}}\right) \text{ (Ans.)}$$

Example-14 If l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of two mutually perpendicular lines then show that $m_1n_2 - m_2n_1$, $n_1l_2 - n_2l_1$, $l_1m_2 - l_2m_1$ be direction cosines of a line perpendicular to both lines.
[পরস্পর লম্ব দুটি সরলরেখার দিক-কোসাইন l_1, m_1, n_1 এবং l_2, m_2, n_2 হলে দেখাও এদের লম্ব সরলরেখার দিক-কোসাইন $m_1n_2 - m_2n_1$, $n_1l_2 - n_2l_1$, $l_1m_2 - l_2m_1$ হবে।]
[RUH '83, CUH '82,

Solution Since l_1, m_1, n_1 and l_2, m_2, n_2 are the direction cosines of two mutually perpendicular lines, then we have

$$l_1^2 + m_1^2 + n_1^2 = 1 \quad \ldots\ldots (1)$$

$$l_2^2 + m_2^2 + n_2^2 = 1 \quad \ldots\ldots (2)$$

and $l_1l_2 + m_1m_2 + n_1n_2 = 0 \quad \ldots\ldots (3)$

Let, l, m, n be the direction cosine of the line which is perpendicular to both lines whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2.

Scanned with CamScanner

Scanned with CamScanner

$\therefore ll_1 + mm_1 + nn_1 = 0$ ..... (4)

and $ll_2 + mm_2 + nn_2 = 0$ ..... (5)

(4) and (5), by cross multiplication, we have

$$\frac{l}{-m_2n_1} = \frac{m}{n_1l_2 - n_2l_1} = \frac{n}{l_1m_2 - l_2m_1} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}}$$

since l, m, n are the direction cosines of a line, then we have

$$l^2 + m^2 + n^2 = 1$$

and $(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2$

$$= (l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1l_2 + m_1m_2 + n_1n_2)^2$$

$$= 1 \cdot 1 - 0 = 1 \quad [\because l^2 + m^2 + n^2 = 1]$$

$$\therefore \frac{l}{m_1n_2 + m_2n_1} = \frac{m}{n_1l_2 - n_2l_1} = \frac{n}{l_1m_2 - l_2m_1} = 1$$

$\therefore l = m_1n_2 - m_2n_1$, $m = n_1l_2 - n_2l_1$ and $n = l_1m_2 - l_2m_1$

efore, the direction cosine of the line, which is perpendicular to both of the s are $m_1n_2 - m_2n_1$, $n_1l_2 - n_2l_1$ and $l_1m_2 - l_2m_1$ [Showed]

) Corresponding formulae when direction of the lines are given
দিকানুপাত দেয়া থাকলে এ সম্পর্কিত সূত্রাবলি

a_1, b_1, c_1 and a_2, b_2, c_2 be the direction ratios of the two given lines. Then their al direction cosines are given by [মনে করি, a_1, b_1, c_1 এবং a_2, b_2, c_2 প্রদত্ত ছয়ের দিকানুপাত। সুতরাং তাদের সঠিক দিক-কোসাইনগুলো]

$$\frac{a_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}}, \frac{b_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}}, \frac{c_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}}$$

and $$\frac{a_2}{\sqrt{(a_2^2 + b_2^2 + c_2^2)}}, \frac{b_2}{\sqrt{(a_2^2 + b_2^2 + c_2^2)}}, \frac{c_2}{\sqrt{(a_2^2 + b_2^2 + c_2^2)}}$$

ng these direction cosines, the angle θ between the two lines is given by [এ কোসাইনগুলো ব্যবহার করে, অন্তর্ভুক্ত কোণ θ পাওয়া যায়।]

$$\cos\theta = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\Sigma a_1^2} \cdot \sqrt{\Sigma a_2^2}}$$

ndition for perpendicularity of two lines (দুটিরেখা পরস্পর লম্ব হওয়ার শর্ত) :

he two lines are perpendicular to each other, then [যদি রেখাদ্বয় পরস্পর লম্ব হয়,

Scanned with CamScanner

Scanned with CamScanner

$$\cos 90° = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\sum a_1^2} \cdot \sqrt{\sum a_2^2}} \quad [\because \theta = 90°]$$

$$\Rightarrow \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\sum a_1^2} \cdot \sqrt{\sum a_2^2}} = 0$$

$$\therefore a_1a_2 + b_1b_2 + c_1c_2 = 0$$

• **Condition of parallelism of two lines (রেখাদ্বয় পরস্পর সমান্তরাল হওয়ার শর্ত)**

If the two lines are parallel to each other, then (যদি রেখাদ্বয় পরস্পর সমান্তরাল হয়,

$$\cos 0° = 1 = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\sum a_1^2} \cdot \sqrt{\sum a_2^2}} \quad [\because \theta = 0°]$$

$$\Rightarrow \left(a_1^2 + b_1^2 + c_1^2\right) \cdot \left(a_2^2 + b_2^2 + c_2^2\right) = (a_1a_2 + b_1b_2 + c_1c_2)^2$$

$$\Rightarrow \left(a_1^2 + b_1^2 + c_1^2\right) \cdot \left(a_2^2 + b_2^2 + c_2^2\right) - (a_1a_2 + b_1b_2 + c_1c_2)^2 = 0$$

$$\therefore (a_1b_2 - a_2b_1)^2 + (b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 = 0$$

The L.H.S. being sum of three squares of real quantities will be zero if ead them is separately zero and hence we have [বামপক্ষে তিনটি সংখ্যার বর্গের যো শূন্য হবে যদি সংখ্যাত্রয় আলাদা আলাদা ভাবে শূন্য হয়, সুতরাং আমরা পাই]

$$a_1b_2 - a_2b_1 = 0,\ b_1c_2 - b_2c_1 = 0,\ c_1a_2 - c_2a_1 = 0$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2},\ \frac{b_1}{b_2} = \frac{c_1}{c_2},\ \frac{c_1}{c_2} = \frac{a_1}{a_2}$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

This shows that the two lines will be parallel if their direction ratios proportional. [দেখা যাচ্ছে যে, দুটি রেখার পরস্পর সমান্তরাল হলে এদের দি কোসাইনগুলো পরস্পর সমানুপাতিক হয়।]

Example-15 Show that, the straight line joining the points (1, 2, –3) to (4, 3 and the straight line joining (4,1, 6) to (1, 2, 7) are perpendicular to each other. [দেখাও যে, (1, 2, –3) ও (4, 3, 5) বিন্দুদ্বয়ের সংযোজক সরলরেখা এবং (4,1, 6) ও (1, বিন্দুদ্বয়ের সংযোগ সরলরেখা পরস্পর লম্ব।]

Solution মনে করি, প্রদত্ত বিন্দুসমূহ A(1, 2, – 3), B(4, 3, 5), C(4 1, 6) এবং D(1, 2 প্রমাণ করতে হবে যে, AB ও CD পরস্পর লম্ব।

AB সরলরেখার দিক-অনুপাত $a_1, b_1, c_1 = 3, 1, 8$ এবং CD সরলরেখার দিক-অনুপাত $b_2, c_2 = -3, 1, 1$

এখন, $a_1a_2 + b_1b_2 + c_1c_2 = 3(-3) + 1.1 + 8.1 = 0$

$\therefore$ AB ও CD পরস্পর লম্ব। [Showed]

Scanned with CamScanner

Area of triangle
ত্রিভুজের ক্ষেত্রফল

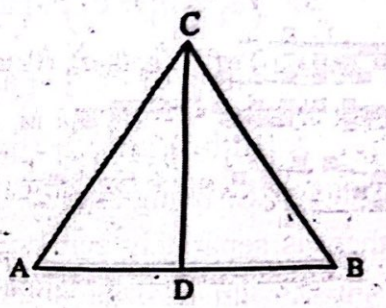
orem-7 : If the vertices of a triangle ABC are $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $y_3, z_3)$ then show that, the area of the triangle is $\Delta = \sqrt{\Delta_x^2 + \Delta_y^2 + \Delta_z^2}$, where-

BC ত্রিভুজের শীর্ষবিন্দুসমূহ $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ এবং $C(x_3, y_3, z_3)$ হলে দেখাও ের ক্ষেত্রফল $\Delta = \sqrt{\Delta_x^2 + \Delta_y^2 + \Delta_z^2}$, যেখানে]

$$\Delta_x = \frac{1}{2}\begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix}, \Delta_y = \frac{1}{2}\begin{vmatrix} z_1 & x_1 & 1 \\ z_2 & x_2 & 1 \\ z_3 & x_3 & 1 \end{vmatrix}, \Delta_z = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

of : মনে করি, ABC ত্রিভুজের AB বাহুর উপর CD লম্ব।

∴ ABC ত্রিভুজের ক্ষেত্রফল,



$$\Delta = \frac{1}{2} \cdot AB \cdot CD \quad \text{.....(1)}$$

বা, ΔACD হতে $\sin A = \frac{CD}{AC}$

$$\Rightarrow CD = AC \sin A$$

)$\Rightarrow \Delta = \frac{1}{2} \cdot AB \cdot AC \sin A$ ..... (2)

ন, AB সরলরেখার দিক-অনুপাত $x_2 - x_1, y_2 - y_1, z_2 - z_1$

∴ AB সরলরেখার দিক-কোসাইন,

$$\frac{x_2 - x_1}{\sqrt{\sum (x_2 - x_1)^2}}, \frac{y_2 - y_1}{\sqrt{\sum (x_2 - x_1)^2}}, \frac{z_2 - z_1}{\sqrt{\sum (x_2 - x_1)^2}}$$

$$\Rightarrow \frac{x_2 - x_1}{AB}, \frac{y_2 - y_1}{AB}, \frac{z_2 - z_1}{AB}$$

রূপভাবে, AC সরলরেখার দিক-কোসাইন,

$$\frac{x_3 - x_1}{\sqrt{\sum (x_3 - x_1)^2}}, \frac{y_3 - y_1}{\sqrt{\sum (x_3 - x_1)^2}}, \frac{z_3 - z_1}{\sqrt{\sum (x_3 - x_1)^2}}$$

$$\Rightarrow \frac{x_3 - x_1}{AC}, \frac{y_3 - y_1}{AC}, \frac{z_3 - z_1}{AC}$$

বা, $\sin^2 A = (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2 m_1)^2$

যে $[l_1, m_1, n_1] = \left[\frac{x_2 - x_1}{AB}, \frac{y_2 - y_1}{AB}, \frac{z_2 - z_1}{AB}\right]$

$[l_2, m_2, n_2] = \left[\frac{x_3 - x_1}{AC}, \frac{y_3 - y_1}{AC}, \frac{z_3 - z_1}{AC}\right]$

Scanned with CamScanner

Scanned with CamScanner

$$\therefore \sin^2 A = \left\{\frac{(y_2-y_1)(z_2-z_1)-(y_3-y_1)(z_2-z_1)}{AB.AC}\right\}^2$$

$$+\left\{\frac{(z_2-z_1)(x_2-x_1)-(z_3-z_1)(x_2-x_1)}{AB.AC}\right\}^2+\left\{\frac{(x_2-x_1)(y_3-y_1)-(x_3-x_1)(y_2-y_1)}{AB.AC}\right\}^2$$

$$\Rightarrow (AB.AC.\sin A)^2 = \begin{vmatrix} y_1 & z_1 & 1 \\ y_2-y_1 & z_2-z_1 & 0 \\ y_3-y_1 & z_3-z_1 & 0 \end{vmatrix}^2 + \begin{vmatrix} z_1 & x_1 & 1 \\ z_2-z_1 & x_2-x_1 & 0 \\ z_3-z_1 & x_3-x_1 & 0 \end{vmatrix}^2$$

$$+\begin{vmatrix} x_1 & y_1 \\ x_2-x_1 & y_2-y_1 \\ x_3-x_1 & y_3-y_1 \end{vmatrix}$$

$$\Rightarrow 4\left(\frac{1}{2}AB\cdot AC.\sin A\right)^2 = 4\left(\Delta_x^2+\Delta_y^2+\Delta_z^2\right)$$

$\therefore \Delta = \sqrt{\Delta_x^2+\Delta_y^2+\Delta_z^2}$ [(2) নং এর সাহায্যে] [Showed]

Example-16 If (a, 0, 0), (0, b, 0) and (0, 0, c) are vertices of a triangle then find its area. [কোন ত্রিভুজের শীর্ষবিন্দুসমূহ (a, 0, 0), (0, b, 0) এবং (0, 0, c) হলে এর ক্ষেত্রফল নির্ণয় কর।]

Solution দেয়া আছে, ত্রিভুজের শীর্ষত্রয় যথাক্রমে (a, 0, 0), (0, b, 0) এবং (0, 0, c)।

$\therefore$ ত্রিভুজের ক্ষেত্রফল, $\Delta = \sqrt{\Delta_x^2+\Delta_y^2+\Delta_z^2}$ ..... (1)

যেখানে, $\Delta_x = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ b & 0 & 1 \\ 0 & c & 1 \end{vmatrix} = \frac{1}{2}bc$

$$\Delta_y = \frac{1}{2}\begin{vmatrix} 0 & a & 1 \\ 0 & 0 & 1 \\ c & 0 & 1 \end{vmatrix} = \frac{1}{2}ca$$

$$\Delta_z = \frac{1}{2}\begin{vmatrix} a & 0 & 1 \\ 0 & b & 1 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{2}ab$$

সুতরাং (1) নং হতে পাই,

$\Delta = \frac{1}{2}\sqrt{b^2c^2+c^2a^2+a^2b^2}$, ইহাই নির্ণেয় ত্রিভুজের ক্ষেত্রফল। (Ans.)

Theorem-8 The perpendicular distance of a point $P(x_1, y_1, z_1)$ from a line through the point A(a, b, c) and whose direction cosines are l, m, n is [$P(x_1, y_1, z_1)$ থেকে l, m, n দিক-কোসাইন বিশিষ্ট এবং A(a, b, c) বিন্দুগামী রেখার লম্ব-দূরত্ব।]

$$PN = \begin{vmatrix} x_1-a & y_1-b \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1-b & z_1-c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1-c & x_1-a \\ n & l \end{vmatrix}^2$$

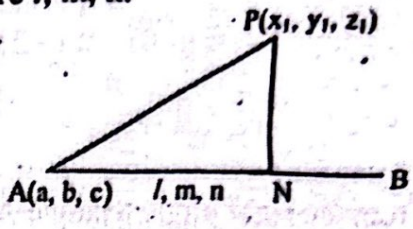
Scanned with CamScanner

Proof : Let AB be a line through A(a, b, c) and whose direction cosines are l, m, n and $P(x_1, y_1, z_1)$ be a point. Draw PN perpendicular to AB from P.

[প্রমাণ : ধরি, A(a, b, c) বিন্দুগামী AB একটি সরলরেখা যার দিক-কোসাইন l, m, n এবং $P(x_1, y_1, z_1)$ একটি বিন্দু। P বিন্দু থেকে AB রেখার উপর PN লম্ব আঁকি।]

Now, AN = Projection of the line segment joining the two points A(a, b, c) and $P(x_1, y_1, z_1)$ on the line AB whose direction cosines are l, m, n.

[এখন, AN = l, m, n দিক-কোসাইন বিশিষ্ট AB রেখার উপর A(a, b, c) এবং $P(x_1, y_1, z_1)$ বিন্দুদ্বয়ের সংযোগ রেখার লম্ব অভিক্ষেপ]



$$\therefore AN = l(x_1 - a) + m(y_1 - b) + n(z_1 + c)$$

And AP = Distance between the two points A(a, b, c) and $P(x_1, y_1, z_1)$ [A(a, b, c) এবং $P(x_1, y_1, z_1)$ বিন্দুদ্বয়ের মধ্যবর্তী দূরত্ব]

$$\therefore AP = \sqrt{\{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2}$$

From the right-angle-triangle APN, we get [APN সমকোণী ত্রিভুজ থেকে আমরা পাই]

$$PN^2 = AP^2 - AN^2$$

$$= \{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2\} - \{l(x_1 - a)^2 + m(y_1 - b) + n(z_1 - c)\}^2$$

$$= \{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2\}\{l^2 + m^2 + n^2) - \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}^2$$

$$= \{l(y_1 - b) - m(x_1 - b)\}^2 + \{m(z_1 - c) - n(y_1 - b)\}^2 + \{n(x_1 - a) - l(z_1 - c)\}^2$$

$$PN = \begin{vmatrix} x_1 - a & y_1 - b \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1 - b & z_1 - c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1 - c & x_1 - a \\ n & l \end{vmatrix}^2$$ [Proved]

Example-17 Find the distance of (– 1, 2, 5) from the line through (3, 4, 5) whose direction cosines are proportional to 2, – 3, 6.

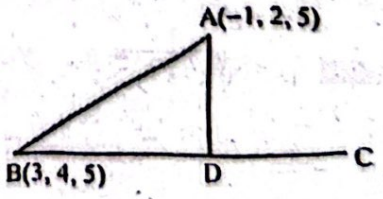
[(3, 4, 5) বিন্দুগামী একটি সরলরেখার দিক-কোসাইনগুলো 2, – 3, 6 এর সমানুপাতিক। (–1, 2, 5) বিন্দু থেকে সরলরেখাটির লম্ব দূরত্ব নির্ণয় কর।]

[NUH 2011, 2012; DUH '93, '95, '99]

Solution Let BC be the line passing through the point B(3, 4, 5) whose direction cosines are proportional to 2, – 3, 6. [ধরি, BC সরলরেখাটি B(3, 4, 5) বিন্দুগামী যার দিক-কোসাইনগুলো 2, – 3, 6 এর সমানুপাতিক।]

Let l, m, n be the direction cosine of BC.

[মনে করি, BC রেখার দিক-কোসাইনগুলো l, m, n.]



$$\frac{l}{2} = \frac{m}{-3} = \frac{n}{6} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{4 + 9 + 36}} = \frac{1}{\sqrt{49}} = \frac{1}{7}$$

$$l = \frac{2}{7}, \qquad m = \frac{-3}{7}, \qquad n = \frac{6}{7}$$

Scanned with CamScanner

Scanned with CamScanner

$\therefore AB = \sqrt{(-1-3)^2 + (2-4)^2 + (5-5)^2} = \sqrt{20} = 2\sqrt{5}$

Draw AD from A perpendicular to BC. [A বিন্দু থেকে BC এর উপর AD লম্ব অঙ্ক

BD = The projection of BA on BC. [BC এর উপর BA এর লম্ব অভিক্ষেপ ।]

$$\therefore BD = \frac{2}{7}(-1-3) - \frac{3}{7}(2-4) + \frac{6}{7}(5-5)$$

$$= \frac{2}{7}(-4) - \frac{3}{7}(-2)$$

$$= \frac{-2}{7}$$

From the right angle triangle ABD, [ABD সমকোণী ত্রিভুজ থেকে পাই,]

$$AD = \sqrt{AB^2 - BD^2}$$

$$= \sqrt{20 - \frac{4}{49}}$$

$$= \sqrt{\frac{976}{49}}$$

$$= \frac{4\sqrt{61}}{7} \text{ (Ans.)}$$

6.12 Polar co-ordinates
পোলার স্থানাঙ্ক

(a) Cylindrical polar co-ordinate (স্তম্ভকীয় পোলার স্থানাঙ্ক) : মনে করি, P বি সমায়ত কার্তেসীয় স্থানাঙ্ক (x, y, z) । P বিন্দু হতে xy সমতলের উপর PN লম্ব হলো । মূলবিন্দু হতে N বিন্দুর দূরত্ব P এমন একটি সিলিন্ডার আঁকি যার কেন্দ্র O ব্যাসার্ধ ON রেখা x অক্ষের সাথে Q কোণ উৎপন্ন করে এবং PN এর দৈর্ঘ্য Z হ বিন্দুর স্তম্ভকীয় (cylindrical) পোলার স্থানাঙ্ক হবে $P(\rho, \varphi, Z)$ ।

কার্তেসীয় স্থানাঙ্ক ও স্তম্ভকীয় স্থানাঙ্কের মধ্যে সম্পর্ক :

চিত্র হতে পাই, $x = OL = ON\cos\varphi = \rho\cos\varphi$

$y = MN = ON\sin\varphi = \rho\sin\varphi$

এবং $z = z$

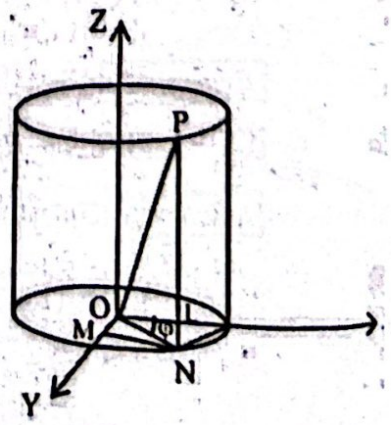
$\therefore x = \rho\cos\varphi$

$y = \rho\sin\varphi$

$z = z$

এবং $\rho = \sqrt{x^2 + y^2}$

$\therefore \varphi = \tan^{-1}\left(\frac{y}{x}\right), z = z$



Some Important Examples

Example-19 Co-ordinates of P and Q are $(1, -5, 7)$, $(-3, 6, -2)$. Fi direction cosines of OP, OQ and PQ.

Solution Let l_1, m_1, n_1 be the direction cosines of OP.

Then $l_1 : m_1 : n_1 = 1 : -5 : 7$

Now $l^2 + (-5)^2 + 7^2 = 75$

$$\therefore l_1 = \frac{1}{\sqrt{75}}, m_1 = \frac{-5}{\sqrt{75}}, n_1 = \frac{7}{\sqrt{75}} \text{ (Ans.)}$$

In the same way, the direction cosines of OQ are found to be

$$-\frac{3}{7}, \frac{6}{7}, -\frac{2}{7} \text{ (Ans.)}$$

Let l, m, n be the diection cosines of PQ.

$$\therefore l : m : n = (-3-1) : (6+5) : (-2-7) = -4 : 11 : -9$$

Since $4^2 + 11^2 + 9^2 = 218$, we have

$$l = \frac{-4}{\sqrt{218}}, m = \frac{11}{\sqrt{218}} \text{ and } n = \frac{-9}{\sqrt{218}} \text{ (Ans.)}$$

Example-20 Determine the perpendicular distance from $(-6, -8, 0)$ to the z axes. [$(-6, -8, 0)$ বিন্দু হতে x, y, z অক্ষের লম্ব দূরত্ব নির্ণয় কর ।]

Solution $(-6, -8, 0)$ বিন্দু হতে x অক্ষের লম্ব দূরত্ব $= \sqrt{(-8)^2 + 0^2} = 8$ (Ans

$(-6, -8, 0)$ বিন্দু হতে y অক্ষের লম্ব দূরত্ব $= \sqrt{(-6)^2 + 0^2} = 6$ (Ans

$(-6, -8, 0)$ বিন্দু হতে z অক্ষের লম্ব দূরত্ব $= \sqrt{(-6)^2 + 8^2} = 10$ (An

Example-21 Find the distance between two points $(2, -1, 3)$ and $(-1, -$ [$(2, -1, 3)$ এবং $(-1, -2, 1)$ বিন্দুদ্বয়ের মধ্যকার দূরত্ব নির্ণয় কর ।]

Solution মনে করি, প্রদত্ত বিন্দুদ্বয় A এবং B.

$$\therefore AB = \sqrt{(2+1)^2 + (-1+2)^2 + (3-1)^2}$$
$$= \sqrt{9+1+4}$$
$$= \sqrt{14} \text{ (Ans.)}$$

Scanned with CamScanner

Ex-22 For what condition three straight lines coincide at one point whose direction ratios are l_1, m_1, n_1, l_2, m_2, n_2 & l_3, m_3, n_3. [কোন শর্তে তিনটি সরলরেখা যাদের দিক-অনুপাত যথাক্রমে l_1, m_1, n_1, l_2, m_2, n_2, l_3, m_3, n_3 এক বিন্দুতে মিলিত হবে?]

Sol মনে করি, l_1, m_1, n_1, l_2, m_2, n_2 দিক-কোসাইন বিশিষ্ট সরলরেখা দুটি একটি তলে অবস্থিত। অপর একটি সরলরেখা যার দিক-কোসাইন l, m, n একই তলের উপর লম্ব। এখন শর্তানুযায়ী,

$ll_1 + mm_1 + nn_1 = 0$ ..... (1)

এবং $ll_2 + mm_2 + nn_2 = 0$ ..... (2)

তলের উপর কোন সরলরেখা লম্ব হলে ইহা ঐ তলের সরলরেখার উপর লম্ব হবে।

সুতরাং ইহা l_3, m_3, n_3 দিক-কোসাইন বিশিষ্ট সরলরেখার উপর লম্ব হবে।

$\therefore ll_3 + mm_3 + nn_3 = 0$ ..... (3)

(1), (2), (3) হতে l, m, n অপসারণ করে পাই,

$$\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$$, ইহাই নির্ণেয় শর্ত। (Ans.)

Ex-23 একটি গতিশীল সরলরেখার দুটি সন্নিহিত অবস্থানের জন্য দিক-কোসাইন l, m, n এবং $(l + \delta l, m + \delta m, n + \delta n)$. দেখাও যে দুটি অবস্থানের জন্য কোণ $\delta\theta$ হলে $(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$.

Sol আমরা জানি,

$\cos\delta\theta = l(l + \delta l) + m(m + \delta m) + n(n + \delta n)$

$\Rightarrow 1 - \cos\delta\theta = -(l\delta l + m\delta m + n\delta n)$ $[\because l^2 + m^2 + n^2 = 1]$

$\Rightarrow 2\sin^2\frac{\delta}{2}\theta = -(l\delta l + m\delta m + n\delta n)$

যেহেতু $\delta\theta$ কোণ ক্ষুদ্র এবং এর প্রথম পদ বিবেচনা করলে,

$2\left(\frac{1}{2}\delta\theta\right)^2 = -(l\delta l + m\delta m + n\delta n)$ ..... (1) $[\because \theta$ ক্ষুদ্র হলে $\sin\theta = \theta]$

এবং $l^2 + m^2 + n^2 = 1$ ..... (2)

$(l + \delta l)^2 + (m + \delta m)^2 + (n + \delta n)^2 = 1$ ..... (3)

(3) হতে (2) বিয়োগ করে পাই,

$(\delta l)^2 + 2l(\delta l) + (\delta m^2) + 2m(\delta m) + (\delta n)^2 + 2n(\delta n) = 0$

$\Rightarrow (\delta l)^2 + (\delta m)^2 + (\delta n)^2 = -2(l\delta l + m\delta m + n\delta n)$

এখন (1) হতে পাই,

$(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$ [Showed]

Scanned with CamScanner

Example-24 Show that, the line joining two points (0, 1, 2) and (3, 4, parallel to the line joining two points (– 4, 3, – 6) and (5, 12, 6).
[দেখাও যে, (0, 1, 2) এবং (3, 4, 6) বিন্দুদ্বয়ের সংযোগ রেখা (– 4, 3, – 6) এবং (5, বিন্দুদ্বয়ের সংযোগ রেখার সাথে পরস্পর সমান্তরাল ।]

Solution The direction ratios of the line joining the two points (0, 1, 2) and (3 are 3 – 0, 4 – 1, 6 – 2. i.e., 3, 3, 4.

Again the diection ratios of the line joining the two points (–4, 3, –6) and (5, 12, 5 + 4, 12 – 3, 6 + 6

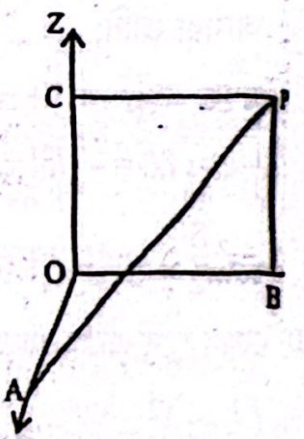
i.e., 9, 9, 12.

We observe that the direction ratios of the two lines are proportional sin

$$\frac{3}{9}=\frac{3}{9}=\frac{4}{12}$$

Hence the given two lines ate parallel. [Showed]

Example-25 Show that, the perpendicular distance from P(x, y, z) to the X-Y-axis and Z-axis are $\sqrt{y^2+z^2}$, $\sqrt{z^2+x^2}$ and $\sqrt{x^2+y^2}$ respectively. Hence fin perpendicular distance from (3, 0, 4) to the X-axis, Y-axis and Z-axis.
[দেখাও যে, P(x, y, z) বিন্দু হতে X-অক্ষ, Y-অক্ষ এবং Z অক্ষের উপর লম্ব দূরত্ব যথ $\sqrt{y^2+z^2}$, $\sqrt{z^2+x^2}$ এবং $\sqrt{x^2+y^2}$; এর সাহায্যে (3, 0, 4) বিন্দু হতে X-অক্ষ, Y-অক্ষ Z অক্ষের উপর লম্ব দূরত্ব নির্ণয় কর ।]



Solution P(x, y, z) বিন্দু হতে X-অক্ষ, Y-অক্ষ এবং Z অক্ষের উপর যথাক্রমে PA, PB এবং PC লম্ব অংকন করি। অক্ষের উপর লম্বগুলোর পাদবিন্দু যথাক্রমে A(x, 0, 0), B(0, y, 0) এবং C(0, 0, z).

এখন P বিন্দু হতে X-অক্ষের উপর লম্ব দূরত্ব,

$$PA=\sqrt{(x-x)^2+(y-0)^2+(z-0)^2}=\sqrt{y^2+z^2}$$

অনুরূপে P বিন্দু হতে Y-অক্ষের উপর লম্ব দূরত্ব $PB=\sqrt{z^2+x^2}$

এবং P বিন্দু হতে Z-অক্ষের উপর লম্ব দূরত্ব $PC=\sqrt{x^2+y^2}$ [Showed]

২য় অংশ : (3, 0, 4) বিন্দু হতে X অক্ষের উপর লম্ব দূরত্ব $=\sqrt{0+16}=4$ (Ans.)

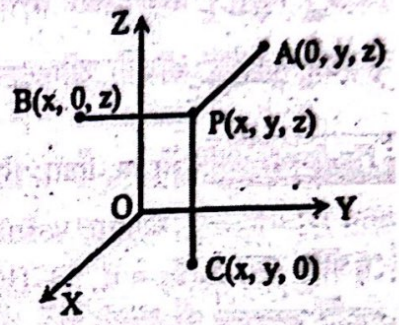
(3, 0, 4) " " Y " " " " $=\sqrt{16+9}=5$ (Ans.)

(3, 0, 4) " " Z " " " " $=\sqrt{9+0}=3$ (Ans.)

Scanned with CamScanner

Example-26 Show that, the perpendicular distance from the point P(x, y, z) to YZ, ZX and XY-plane are |x|, |y| and |z| respectively. Hence find the perpendicular distance from (1, – 2, –3) to YZ, ZX and XY-plane.
দেখাও যে, P(x, y, z) বিন্দু হতে YZ, ZX এবং XY-সমতলের উপর লম্ব দূরত্বগুলো যথাক্রমে |x|, |y| এবং |z| ; এর সাহায্যে (1, – 2, – 3) বিন্দু হতে YZ, ZX এবং XY-সমতল উপর লম্ব দূরত্ব নির্ণয় কর ।]

Solution P(x, y, z) বিন্দু হতে YZ, ZX এবং XY সমতলের উপর PA, PB এবং PC অঙ্কন করি; যেখানে A, B, C বিন্দুগুলো লম্বের পাদবিন্দু ।



সুতরাং A, B, C বিন্দুগুলোর স্থানাঙ্ক যথাক্রমে (0, y, z), (x, 0, z) এবং (x, y, 0).

এখন P বিন্দু হতে YZ সমতলের লম্ব দূরত্ব,

$$PA = \sqrt{(x-0)^2 + (y-y)^2 + (z-z)^2} = \sqrt{x^2} = |x|$$

অনুরূপভাবে, P বিন্দু হতে ZX সমতলের লম্ব দূরত্ব $PB = \sqrt{y^2} = |y|$

এবং P বিন্দু হতে XY সমতলের লম্ব দূরত্ব $PC = \sqrt{z^2} = |z|$ [Showed]

নির্ণেয় : (1, – 2, – 3) বিন্দু হতে YZ সমতলের লম্ব দূরত্ব = | 1 | = 1 (Ans.)

(1, – 2, – 3) বিন্দু হতে ZX সমতলের লম্ব দূরত্ব = | – 2 | = 2 (Ans.)

(1, – 2, – 3) বিন্দু হতে YZ সমতলের লম্ব দূরত্ব = | – 3 | = 3 (Ans.)

Example-27 Show that, the triangle formed by the points A(1, – 2, – 3), B(2, – 3, – 1) and C(3, – 1, – 2) is an equilateral triangle.
দেখাও যে, A(1, – 2, – 3), B(2, – 3, – 1) এবং C(3, – 1, – 2) বিন্দুগুলো দ্বারা একটি সমবাহু ত্রিভুজ গঠিত হয় ।]

Solution এখানে, $AB = \sqrt{(2-1)^2 + (-3+2)^2 + (-1+3)^2} = \sqrt{(1+1+4)} = \sqrt{6}$

$$BC = \sqrt{(3-1)^2 + (-1+2)^2 + (-2+1)^2} = \sqrt{(4+1+1)} = \sqrt{6}$$

$$AC = \sqrt{(3-1)^2 + (-1+2)^2 + (-2+3)^2} = \sqrt{(4+1+1)} = \sqrt{6}$$

যেহেতু AB = BC = AC, সুতরাং ABC একটি সমবাহু ত্রিভুজ । [Showed]

Example-28 Show that, the four points A(4, 2, – 6), B(5, – 3, 1), C(12, 4, 5) and D(11, 9, – 2) are vertices of a rectangle.
দেখাও যে, A(4, 2, – 6), B(5, – 3, 1), C(12, 4, 5) এবং D(11, 9, – 2) বিন্দু চারটি একটি আয়তক্ষেত্রের শীর্ষবিন্দু ।]

Solution এখানে, $AB = \sqrt{(1+25+49)} = 5\sqrt{3}$

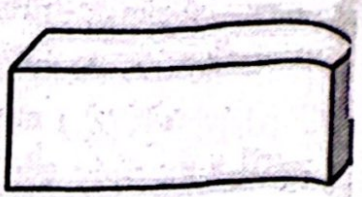
$BC = \sqrt{(49+49+16)} = \sqrt{114}$

$CD = \sqrt{(1+25+49)} = 5\sqrt{3}$

$AD = \sqrt{(49+49+16)} = \sqrt{114}$

$AC = \sqrt{(64+4+121)} = 3\sqrt{21}$

$BD = \sqrt{(36+144+9)} = 3\sqrt{21}$



যেহেতু ABCD চতুর্ভুজের বিপরীত বাহুদ্বয় AB = DC, BC = AD এবং কর্ণদ্বয় AC = B

সুতরাং ABCD একটি আয়তক্ষেত্র। [Showed]

Example-29 Show that, the four points A(5, – 1, 1), B(7, – 4, 7), C(1, – 6, and D(– 1, – 3, 4) are vertices of a rombus.
[দেখাও যে, A(5, – 1, 1), B(7, – 4, 7), C(1, – 6, 10) এবং D(– 1, – 3, 4) বিন্দু চ একটি রম্বসের শীর্ষবিন্দু।]

Solution From given points we get,

$AB = \sqrt{4+9+36} = \sqrt{49} = 7$

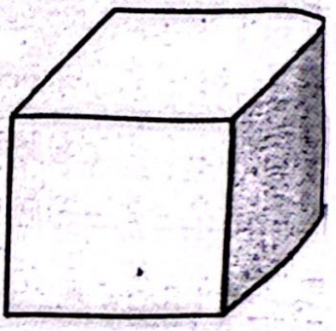
$BC = \sqrt{36+4+9} = \sqrt{49} = 7$

$CD = \sqrt{4+9+36} = \sqrt{49} = 7$

$AD = \sqrt{36+4+9} = \sqrt{49} = 7$

$AC = \sqrt{16+25+81} = \sqrt{122}$

$BD = \sqrt{64+1+9} = \sqrt{74}$



যেহেতু ABCD চতুর্ভুজের চারটি বাহু AB = BC = CD = AD এবং কর্ণদ্বয় AC ≠ BD

সুতরাং ABCD একটি রম্বস। [Showed]

Example-30 Find the projection of the join of points (– 1, – 1, 3) and (2, 0, 1) the line through the points (– 7, 5, 3) and (2, 6, 8).

Solution Let l, m, n be the direction cosines of the line through the points (– 7, 3) and (2, 6, 8). Then,

$$\frac{l}{-7-2} = \frac{m}{5-6} = \frac{n}{3-8}$$

$$\text{or, } \frac{l}{9} = \frac{m}{1} = \frac{n}{5} = \sqrt{\frac{l^2+m^2+n^2}{9^2+1^2+5^2}} = \frac{1}{\sqrt{107}} \quad [\because l^2+m^2+n^2 = 1]$$

Scanned with CamScanner

ace, $l=\frac{9}{\sqrt{107}}, m=\frac{1}{\sqrt{107}}, n=\frac{5}{\sqrt{107}}$ ..... (1)

e projection of the join of points (– 1, – 1, 3) and (2, 0, 1) on the given line is

$l\{2-(-1)\}+m\{0-(-1)\}+n(1-3)$

$=3l+m-2n=\frac{1}{\sqrt{107}}(3\times 9+1-2\times 5)$ [using (1)]

$=\frac{18}{\sqrt{107}}$ (Ans.)

mple-31 Show that, the four points (2, –1, 0) (0, – 1, –1), (1, 1, – 3) and (3, 1, –2) rtices of a rectangle.

ণ যে, (2, – 1, 0), (0, – 1, – 1) (1, 1, – 3) এবং (3, 1, – 2) বিন্দু চারটি একটি ক্ষেত্রের শীর্ষ বিন্দু ।]

ution মনে করি, প্রদত্ত বিন্দুগুলো ABCD ক্ষেত্রের শীর্ষবিন্দু; তাহলে–

$AB=\sqrt{(2-0)^2+(-1+1)^2+(0+1)^2}=\sqrt{5}$

$BC=\sqrt{(0-1)^2+(-1-1)^2+(-1+3)^2}=\sqrt{9}$

$CD=\sqrt{(1-3)^2+(1-1)^2+(-3+2)^2}=\sqrt{5}$

$DA=\sqrt{(3-2)^2+(1-1)^2+(-2-0)^2}=\sqrt{9}$

$AC=\sqrt{(2-1)^2+(-1-1)^2+(-0+3)^2}=\sqrt{14}$

$BD=\sqrt{(0-3)^2+(-1-1)^2+(-1+2)^2}=\sqrt{14}$

ন বিপরীত বাহু, AB = CD এবং BC = DA

র কর্ণদ্বয়, AC = BD

, B, C, D বিন্দু চারটি একটি আয়ক্ষেত্রের শীর্ষবিন্দু । [Showed]

mple-32 Find the direction cosines of a straight line which is equally inclined e axes. If each of the angles be α then show that, $\sin\alpha=\pm\sqrt{2/3}$.

সরলরেখা অক্ষত্রয়ের সাথে সমান কোণ উৎপন্ন করে তার দিক-কোসাইন নির্ণয় কর। গুলোর প্রত্যেকটি α হলে দেখাও যে, $\sin\alpha=\pm\sqrt{2/3}$.] *[DUH '93; RUH '82; H '85]*

ution মনে করি, সরলরেখাটি অক্ষত্রয়ের সাথে সমান কোণ α উৎপন্ন করে।

ল, সরলরেখাটির দিক-কোসাইন : $\cos\alpha, \cos\alpha, \cos\alpha$.

া জানি, $\cos^2\alpha+\cos^2\alpha+\cos^2\alpha=1$

$\Rightarrow 3\cos^2\alpha=1$

Scanned with CamScanner

$\Rightarrow \cos\alpha = \pm\frac{1}{\sqrt{3}}$ ..... (1)

সুতরাং অক্ষত্রয়ের সাথে সমান কোণে আনত সরলরেখার দিক-কোসাইন,

$\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ অথবা, $-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$ (Ans.)

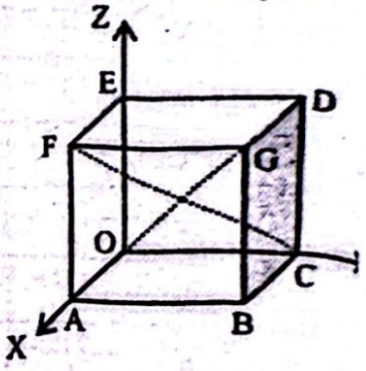
আবার, (1) $\Rightarrow 1 - \sin^2\alpha = \frac{1}{3} \Rightarrow \sin\alpha = \pm\sqrt{\frac{2}{3}}$ [Showed]

Example-33 Prove that, the angle between two diagonals of a cube is $\cos^{-1}\left(\frac{1}{3}\right)$

[দেখাও যে, একটি ঘনকের দুটি কর্ণের অন্তর্গত কোণ $\cos^{-1}\left(\frac{1}{3}\right)$]

[NUH 2001, 2012, 2014, 2018; NUH(NM) '02,]

Solution মনে করি, a ধারের দৈর্ঘ্য বিশিষ্ট OABCDEFG একটি ঘনক যার মূলবিন্দু এবং OA, OC ও OE ধারসমূহ যথাক্রমে X-অক্ষ, Y-অক্ষ ও Z-অক্ষ বরাবর অবস্থিত।



∴ O বিন্দুর স্থানাঙ্ক : (0, 0, 0) A বিন্দুর স্থানাঙ্ক (a, 0, 0)

B বিন্দুর স্থানাঙ্ক : (a, a, 0) C বিন্দুর স্থানাঙ্ক (0, a, 0)

D বিন্দুর স্থানাঙ্ক : (0, a, a) E বিন্দুর স্থানাঙ্ক (0, 0, a)

F বিন্দুর স্থানাঙ্ক : (a, 0, a) G বিন্দুর স্থানাঙ্ক (a, a, a)

ঘনকের দুটি কর্ণ OG এবং CF. এখন OG কর্ণের দিক-অনুপাত : a, a, a

অর্থাৎ 1, 1, 1; একইভাবে CF কর্ণের দিক-অনুপাত : 1, – 1, 1

সুতরাং কর্ণদ্বয়ের মধ্যবর্তী কোণ θ হলে $\cos\theta = \left(\frac{1.1 - 1.1 + 1.1}{\sqrt{1+1+1}\sqrt{1+1+1}}\right)$

$\Rightarrow \theta = \cos^{-1}\left(\frac{1.1 - 1.1 + 1.1}{\sqrt{1+1+1}\sqrt{1+1+1}}\right)$

$= \cos^{-1}\left(\frac{1}{3}\right)$ [Proved]

Example-34 If A(1, 2, 3), B(5, 0, – 1) and C(3, – 2, 5) then show that $EF = \frac{1}{2}$

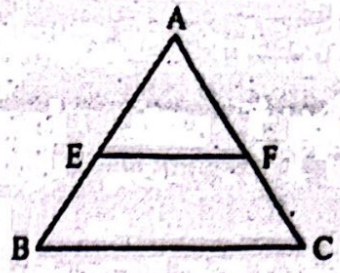
where E and F are middle points of AB and AC respectively.

[যদি A(1, 2, 3), B(5, 0, – 1) এবং C(3, – 2, 5) হয় তবে দেখাও যে, $EF = \frac{1}{2}BC$. যে

E এবং F যথাক্রমে AB এবং AC এর মধ্যবিন্দু।]

Scanned with CamScanner

Solution ... ABC ত্রিভুজের শীর্ষ গুলো যথাক্রমে $A(1, 2, 3)$, $B(5, 0, -1)$ এবং $C(3, -2, 5)$.



এখানে, E বিন্দুর স্থানাঙ্ক $= \left(\frac{1+5}{2}, \frac{2+0}{2}, \frac{3-1}{2}\right) = (3, 1, 1)$

F বিন্দুর স্থানাঙ্ক $= \left(\frac{1+3}{2}, \frac{2-2}{2}, \frac{3+5}{2}\right) = (2, 0, 4)$

$\therefore EF = \sqrt{(1+1+9)} = \sqrt{11}$ এবং $BC = \sqrt{(4+4+36)} = \sqrt{44} = 2\sqrt{11}$

$\Rightarrow \frac{1}{2}BC = \sqrt{11} \Rightarrow EF = \frac{1}{2}BC$ [Showed]

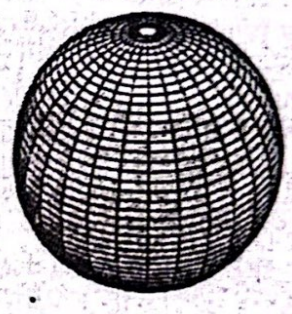
Example-35 Find the locus of those points whose distance from (1, 2, 2) is 4.
[(1, 2, 2) বিন্দু হতে 4 একক দূরত্বের বিন্দুসমূহের সঞ্চারপথের সমীকরণ নির্ণয় কর।]

Solution মনে করি, (x, y, z) বিন্দু হতে (1, 2, 2) বিন্দুর দূরত্ব 4 একক।

সুতরাং আমরা পাই, $\sqrt{(x-1)^2 + (y-2)^2 + (z-2)^2} = 4$

$\Rightarrow \sqrt{(x-1)^2 + (y-2)^2 + (z-2)^2} = 4$

$\Rightarrow x^2 + y^2 + z^2 - 2x - 4y - 4z + 5 = 0$



ইহাই নির্ণেয় সঞ্চার পথের সমীকরণ। (Ans.)

Example-36 If A, B, C, D be the four points whose co-ordinates are respectively (–1, 1, 0), (3, –3, 6), (2, –1, 2) and (4, 7, 6). Show that, AB is perpendicular to CD.

Solution Let a_1, b_1, c_1 be the direction ratios of the line AB and a_2, b_2, c_2 be those of the line CD. Then,

$a_1 = 3 - (-1) = 4, b_1 = -3 - 1 = -4, c_1 = 6 - 0 = 6$

and $a_2 = 4 - 2 = 2, b_2 = 7 - (-1) = 8, c_2 = 6 - 2 = 4$

Now AB will be perpendicular to CD if

$a_1a_2 + b_1b_2 + c_1c_2 = 0$

or, if $4 \times 2 + (-4) \times 8 + 6 \times 4 = 0$

Which is clearly true. Hence the result. [Showed]

Example-37 Show that, the straight line joining points (2, 3, – 2) to (3, 1, 1) and straight line joining (2, 1, – 5) to (4, – 3, 1) are parallel to each other.
[দেখাও যে, (2, 3, – 2) ও (3, 1, 1) বিন্দুদ্বয়ের সংযোজক সরলরেখা এবং (2, 1, – 5) ও (4, –3, 1) বিন্দুদ্বয়ের সংযোগ সরলরেখা পরস্পর সমান্তরাল।]

Scanned with CamScanner

Solution মনে করি, প্রদত্ত বিন্দুসমূহ A(2, 3, – 2), B(3, 1, 1), C(2, 1, – 5) এবং D(4, 3, 1) । প্রমাণ করতে হবে AB ও CD পরস্পর সমান্তরাল ।

AB সরলরেখার দিক-অনুপাত, $a_1, b_1, c_1 = 3-2, 1-3, 1+2$
$= 1, -2, 3$

CD সরলরেখার দিক-অনুপাত, $a_2, b_2, c_2 = 4-2, -3-1, 1+5$
$= 2, -4, 6$

এখন $\frac{a_1}{a_2} = \frac{1}{2}, \frac{b_1}{b_2} = \frac{1}{2}, \frac{c_1}{c_2} = \frac{1}{2}$

অর্থাৎ $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

∴ AB ও CD পরস্পর সমান্তরাল । [Showed]

Example-38 Prove that, the two straight lines whose direction cosines are given by two relations $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$ are perpendicular or parallel according as $a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$ or, $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$.

দুটি সরলরেখার দিক-কোসাইন $al + bm + cn = 0$ এবং $ul^2 + vm^2 + wn^2 = 0$ সম্পর্ক দ্বারা প্রকাশিত হলে প্রমাণ কর যে, সরলরেখা দুটি পরস্পর লম্ব অথবা সমান্তরাল হবে যথাক্রমে $a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$ অথবা, $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$ শর্ত সাপেক্ষে ।]

[NUH 1993, 2010, 2013; CUH '86, '83; DUH '04; RUH '85]

Solution The given two relations are

$$al + bm + cn = 0 \quad \ldots\ldots (1)$$
$$\text{and } ul^2 + vm^2 + wn^2 = 0 \quad \ldots\ldots (2)$$

From the relation (1), we get

$$cn = -(al + bm)$$
$$n = -\left(\frac{al + bm}{c}\right)$$

Putting the value of n in relation (2), we get

$$ul^2 + vm^2 + w\left\{-\left(\frac{al + bm}{c}\right)\right\}^2 = 0$$
$$\Rightarrow uc^2l^2 + vc^2m^2 + w(a^2l^2 + 2ablm + b^2m^2) = 0$$
$$\Rightarrow (uc^2 + wa^2)l^2 + 2wablm + (wb^2 + vc^2)m^2 = 0$$
$$\therefore (uc^2 + wa^2)\left(\frac{l}{m}\right)^2 + 2wab\left(\frac{l}{m}\right) + (wb^2 + vc^2) = 0 \quad \ldots\ldots (3)$$

Scanned with CamScanner

Let l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of the given two lines.

Equation (3) is a quadratic equation of $\left(\frac{l}{m}\right)$. So it has two roots. Let $\frac{l_1}{m_1}$ and $\frac{l_2}{m_2}$ be the two roots of equation (3), then the product of the roots are $\left(\frac{l_1}{m_1}\right).\left(\frac{l_2}{m_2}\right) = \frac{wb^2 + vc^2}{uc^2 + wa^2}$

$$\therefore \frac{l_1 l_2}{wb^2 + vc^2} = \frac{m_1 m_2}{uc^2 + wa^2} \quad \ldots\ldots\ldots\ldots (4)$$

Similarly, eliminating l from relation (1) and (2) we get

$$\frac{m_1 m_2}{uc^2 + wa^2} = \frac{n_1 n_2}{a^2 v + b^2 u} \quad \ldots\ldots\ldots\ldots (5)$$

From (4) and (5) we get $\frac{l_1 l_2}{wb^2 + vc^2} = \frac{m_1 m_2}{uc^2 + wa^2} = \frac{n_1 n_2}{a^2 b + b^2 u} = k$ (say)

$\therefore l_1 l_2 = k(wb^2 + vc^2)$, $m_1 m_2 = k(uc^2 + wa^2)$ and $n_1 n_2 = k(a^2 v + b^2 u)$

- The two lines will be perpendicular if

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$\Rightarrow k(wb^2 + vc^2) + k(uc^2 + wa^2) + k(a^2 v + b^2 u) = 0$$

$$\therefore a^2(v + w) + b^2(w + u) + c^2(u + v) = 0 \text{ [Proved]}$$

- The two lines will be parallel if two roots of equation (3) are equal. Hence the discriminant of equation (3) is equal to zero.

$$(2wab)^2 - 4(uc^2 + wa^2).(wb^2 + vc^2) = 0$$

$$\Rightarrow 4w^2a^2b^2 - 4wub^2c^2 - 4uvc^4 - 4w^2a^2b^2 - 4vwc^2a^2 = 0$$

$$\Rightarrow wub^2c^2 + uvc^4 + vwc^2a^2 = 0$$

Dividing both sides by $uvwc^2$, we get

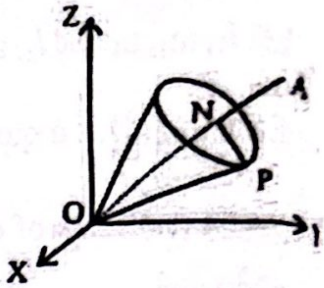
$$\therefore \frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0 \text{ [Proved]}$$

Example-39 Show that, the equation to the right circular cone whose vertex is at the origin, axis has direction cosines l, m, n and semi-vertical angle θ is $(mx - ly)^2 + (ny - mz)^2 + (lz - nx)^2 = (x^2 + y^2 + z^2)\sin^2\theta$.

[যে সমবৃত্তীয় কোণকের অক্ষের দিক-কোসাইন l, m, n; শীর্ষবিন্দু মূলবিন্দুতে এবং অর্ধশীর্ষকোণ θ দেখাও যে, তার সমীকরণ $(mx - ly)^2 + (ny - mz)^2 + (lz - nx)^2 = (x^2 + y^2 + z^2)\sin\theta$.]

Solution মনে করি, কোণকের উপরস্থ যে কোন বিন্দু P(x, y, z). P হতে কোণকের অক্ষ OA এর উপর PN লম্ব অঙ্কন করি।

$\therefore$ ΔOPN হতে পাই $PN^2 = OP^2 - ON^2$ ..... (1)



এবং $PN = OP \sin\theta$

এখানে, $OP^2 = x^2 + y^2 + z^2$

$\Rightarrow PN^2 = (x^2 + y^2 + z^2)\sin^2\theta$

আবার, $ON = l, m, n$ দিক-কোসাইন বিশিষ্ট রেখার উপর OP এর লম্ব অভিক্ষেপ,

$= l(x-0) + m(y-0) + n(z-0) = lx + my + nz$

$(1) \Rightarrow (x^2 + y^2 + z^2)\sin^2\theta = (x^2 + y^2 + z^2) - (lx + my + nz)^2 = (l^2 + m^2 + n^2)(x^2 + y^2 + z^2) - (lx + my + nz)^2$

$\therefore (x^2 + y^2 + z^2)\sin^2\theta = (mx - ly)^2 + (ny - mz)^2 + (lz - nx)^2$

ইহাই নির্ণেয় সমবৃত্তীয় কোণকের সমীকরণ। [Showed]

Example-40 A line through the origin makes angles α, β, γ with its projections on the co-ordinate planes. The distance of any point (x, y, z) from the line and its projections are d and a, b, c prove that $d^2 = (a^2 - x^2)\cos^2\alpha + (b^2 - y^2)\cos^2\beta + (c^2 - z^2)\cos^2\gamma$.

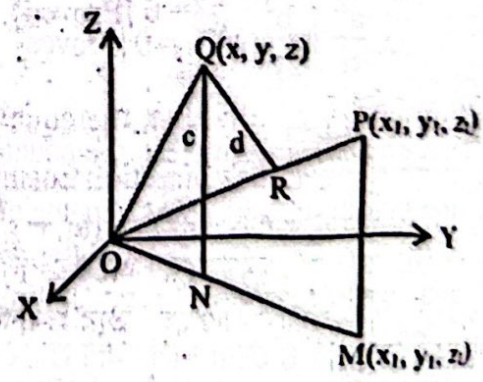
মূল বিন্দুগামী কোন সরলরেখার স্থানাঙ্ক অক্ষ সমতলসমূহে ইহার অভিক্ষেপের সাথে α, β, γ কোণ উৎপন্ন করে। ঐ সরলরেখাটি এবং ইহার অভিক্ষেপগুলো হতে কোন একটি বিন্দু (x, y, z) এর দূরত্ব যথাক্রমে d এবং a, b, c হলে প্রমাণ কর যে, $d^2 = (a^2 - x^2)\cos^2\alpha + (b^2 - y^2)\cos^2\beta + (c^2 - z^2)\cos^2\gamma$] *[DUH '87; RUH '83, '85, '93; NUH '93]*

Solution Let OP be the line which passes through the origin and makes angles α, β, γ with its projections on the co-ordiante planes yz, zx and xy respectively. Draw PM perpendicular to xy plane from the point $P(x_1, y_1, z_1)$. The foot of the perpendicular is $M(x_1, y_1, 0)$. OM is projection of the line OP on the xy plane. So $\angle POM = \gamma$.

$\therefore OM = OP\cos\gamma$

$\Rightarrow \sqrt{(x_1^2 + y_1^2)} = \sqrt{(x_1^2 + y_1^2 + z_1^2)}\cos\gamma$

$\therefore \cos^2\gamma = \dfrac{x_1^2 + y_1^2}{x_1^2 + y_1^2 + z_1^2}$ ..... (1)



Similarly, draw perpendicular to yz and zx plane from the point $p(x_1, y_1, z_1)$, we get

$\cos^2\alpha = \dfrac{y_1^2 + z_1^2}{x_1^2 + y_1^2 + z_1^2}$ ..... (2)

and, $\cos^2\beta = \dfrac{z_1^2 + x_1^2}{x_1^2 + y_1^2 + z_1^2}$ ..... (3)

Scanned with CamScanner

Scanned with CamScanner

Q(x, y, z) be any point. The distance of the point Q(x, y, z) from the line OP is d its projections are a, b, c respectively. Draw perpendicular QR and QN from x, y, z) to OP and OM respectively. Hence QR = d and QN = c.

direction ratios of the line OP are x_1, y_1, z_1. So the direction cosines of the OP.

$$\frac{x_1}{\sqrt{(x_1^2+y_1^2+z_1^2)}}, \frac{y_1}{\sqrt{(x_1^2+y_1^2+z_1^2)}}, \frac{z_1}{\sqrt{(x_1^2+y_1^2+z_1^2)}}$$

w OR = The projection of OQ on the line OP.

$$= \frac{x x_1 + y y_1 + z z_1}{\sqrt{(x_1^2+y_1^2+z_1^2)}}$$

m the right-angle triangle OQR, we have $QR^2 = OQ^2 - OR^2$

$$\Rightarrow d^2 = (x^2+y^2+z^2) - \frac{(xx_1+yy_1+zz_1)^2}{x_1^2+y_1^2+z_1^2}$$

$$\Rightarrow d^2 = \frac{(x^2+y^2+z^2)(x_1^2+y_1^2+z_1^2) + (xx_1+yy_1+zz_1)^2}{x_1^2+y_1^2+z_1^2}$$

$$\therefore d^2 = \frac{(xy_1-x_1y)^2 + (yz_1-y_1z)^2 + (zx_1-z_1x)^2}{x_1^2+y_1^2+z_1^2} \quad \ldots\ldots (4)$$

[By Lagrange identity]

ilarly, putting $z_1 = 0$ in (4)

$$c^2 = \frac{(xy_1-x_1y)^2 + (y.0-y_1z)^2 + (ax_1-0.x)^2}{x_1^2+y_1^2+0}$$

$$\Rightarrow c^2 = \frac{(xy_1-x_1y)^2 + y_1^2z^2 + z^2x_1^2}{x_1^2+y_1^2}$$

$$\Rightarrow c^2 = \frac{(xy_1-x_1y)^2 + (x_1^2+y_1^2)z^2}{x_1^2+y_1^2}$$

$$\Rightarrow c^2 = z^2 + \frac{(xy_1-x_1y)^2}{x_1^2+y_1^2}$$

$$\therefore c^2 - z^2 = \frac{(xy_1-x_1y)^2}{x_1^2+y_1^2} \quad \ldots\ldots (5)$$

ltiplying (1) and (5), we get

$$(c^2-z^2)\cos^2\gamma = \frac{(xy_1-x_1y)^2}{x_1^2+y_1^2+z_1^2} \quad \ldots\ldots (6)$$

ilarly, $(a^2-x^2)\cos^2\alpha = \frac{(yz_1-y_1z)^2}{x_1^2+y_1^2+z_1^2} \quad \ldots\ldots (7)$

Scanned with CamScanner

and $(b^2 - y^2)\cos^2\beta = \frac{(zx_1 - z_1x)^2}{x_1^2 + y_1^2 + z_1^2}$ ..... (8)

Adding (6), (7) and (8) and with the help of (4), we get

$d^2 = (a^2 - x^2)\cos^2\alpha + (b^2 - y^2)\cos^2\beta + (c^2 - z^2)\cos^2\gamma$ [Proved]

Example-41 A, B, C, D are four points in space such that AB ⊥ CD and AC ⊥ BD then show that, AD ⊥ BC.

[যদি Space এ অবস্থিত চারটি বিন্দু A, B, C, D এরূপ যে AB ⊥ CD এবং AC ⊥ BD তবে দেখাও যে, AD ⊥ BC.] *[RUH]*

Solution মনে করি, A, B, C, D বিন্দুগুলোর স্থানাঙ্ক যথাক্রমে (0, 0, 0), (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3).

এখন, AB এর দিক-অনুপাত, $x_1 - 0, y_1 - 0, z_1 - 0$

CD এর দিক-অনুপাত, $x_3 - x_2, y_3 - y_2, z_3 - z_2$

AC এর দিক-অনুপাত, $x_2 - 0, y_2 - 0, z_2 - 0$

BD এর দিক-অনুপাত, $x_3 - x_1, y_3 - y_1, z_3 - z_1$

যেহেতু AB ⊥ CD ∴ $x_1(x_3 - x_2) + y_1(y_3 - y_2) + z_1(z_3 - z_2) = 0$ ..... (1

আবার, AC ⊥ BD ∴ $x_2(x_3 - x_1) + y_2(y_3 - y_1) + z_2(z_3 - z_1) = 0$ ..... (2

(1) নং থেকে (2) নং বিয়োগ করে পাই,

$x_3(x_1 - x_2) + y_3(y_1 - y_2) + z_3(z_1 - z_2) = 0$ ..... (3)

যেহেতু AD এর দিক-অনুপাত, $x_3 - 0, y_3 - 0, z_3 - 0$

এবং BC এর দিক-অনুপাত, $x_1 - x_2, y_1 - y_2, z_1 - z_2$

সুতরাং (3) নির্দেশ করে AD ⊥ BC. [Showed]

Example-42 Find the equation of the right circular cone whose vertex is (2, – semi-vertical angle 30° and whose axis makes equal angle with the co-ordi axes.

[(2, – 3, 5) শীর্ষ বিন্দু বিশিষ্ট একটি সমবৃত্তাকার কোণকের সমীকরণ নির্ণয় কর যার স্থানাঙ্ক অক্ষদলের সাথে সমান কোণ উৎপন্ন করে এবং যার অর্ধশীর্ষ কোণ 30°] *[CUH]*

Solution Let P(2, – 3, 5) be the vertex of the right circular cone whose axis makes equal angle with the co-ordinate axes. So the direction ratios of the line are 1, 1, 1.

Let R(x, y, z) be any point on the right circular cone. It is given that the semi vertical angle ∠RPQ = 30°

Scanned with CamScanner

Scanned with CamScanner

Example-44 $(-9, 4, 5)$ ও $(11, 0, -1)$ বিন্দুদ্বয়ের সংযোগ সরলরেখার উপর ম হতে অঙ্কিত লম্ব সরলরেখা যে বিন্দুতে মিলিত হয় তা নির্ণয় কর।

Solution মনে করি, $A(11, 0, -1)$ এবং $B(-9, 4, 5)$ বিন্দুদ্বয়ের সংযোগ সরলরেখার উপর মূলবিন্দু $O(0, 0, 0)$ হতে OC লম্ব অঙ্কন করা হলো; যেখানে লম্বের পাদবিন্দু C যদি AB সরলরে বিন্দুতে $k : 1$ অনুপাতে অন্তঃবিভক্ত হয় তবে C বিন্দুর স্থানাঙ্ক $\left(\frac{-9k+11}{k+1}, \frac{4k}{k+1}, \frac{5k-1}{k+1}\right)$

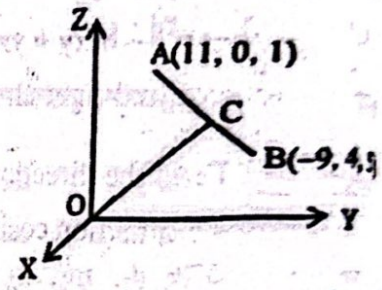
এখন OC সরলরেখার দিক-অনুপাত $-9k + 11, 4k, 5k - 1$ এবং AB সরলরেখার অনুপাত $-20, 4, 6$. যেহেতু $OC \perp AB$;

সুতরাং $-20(-9k+11) + 4(4k) + 6(5k-1) = 0$

$\Rightarrow 90k - 110 + 8k + 15k - 3 = 0$

$\Rightarrow 113k = 113 \Rightarrow k = 1$

সুতরাং C বিন্দুর নির্ণেয় স্থানাঙ্ক $(1, 2, 2)$ (Ans.)



Shortest Questions with Answers

1. What is the distance between the points (x_1, y_1, z_1) and (x_2, y_2, z_2)? [যে দুটি বিন্দুর স্থানাঙ্ক (x_1, y_1, z_1) ও (x_2, y_2, z_2) হলে বিন্দু দুটির মধ্যবর্তী দূরত্ব কত]

Ans. $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$.

2. What is co-ordinate of p, which internally divided the line joining (x_1, z_1) and (x_2, y_2, z_2) in $m_1 : m_2$? [(x_1, y_1, z_1) ও (x_2, y_2, z_2) বিন্দু দ্বয় সংযোগকারী সরলরেখাটি $p(x, y, z)$ বিন্দু দ্বারা $m_1 : m_2$ অনুপাতে অন্তঃবিভক্ত হয় বিন্দুর স্থানাঙ্ক কত?]

Ans. $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}, \frac{m_1z_2 + m_2z_1}{m_1 + m_2}\right)$

3. What is called direction cosine? [দিক-কোসাইন কাকে বলে?]

[NUH 2010, 2013, 2016, 20

Ans. কোন সরলরেখা স্থানাঙ্কের অক্ষগুলোর ধনাত্মক দিকের সাথে যে কোণগুলো উৎ করে তাদের কোসাইনকে সরলরেখাটির দিক-কোসাইন বলে।

4. What is the projection of a straight line joining the points (x_1, y_1, z_1) (x_2, y_2, z_2) on a straight line whose direction cosines are l, m, n? [l, m দিক-কোসাইন বিশিষ্ট রেখার উপর (x_1, y_1, z_1) ও (x_2, y_2, z_2) বিন্দুদ্বয়ের সংযো রেখার লম্ব অভিক্ষেপ কত?]

Ans. প্রদত্ত শর্তে লম্ব অভিক্ষেপ : $l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$.

5. What is the direction cosine of X axis? [X-অক্ষের দিক-কোসাইন কত?]

[NUH 20

Ans. $\cos \alpha$.